The AI Paradox:

How Optimization May Undermine Meaningful Work in Helping Professions

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Abstract

Helping professions are adopting artificial intelligence (AI) to automate documentation, monitoring, and coordination on the premise that offloading routine tasks frees workers for meaningful care. That premise assumes managers know which tasks are routine and which meaningful. We argue instead for a task-contingent AI paradox: meaning in helping work concentrates in identity-central tasks that are increasingly automatable. Study 1, a two-wave critical-incident panel of 307 helping-profession workers (144 longitudinal cases one year later), tested six pre-specified predictions that Time 1 human-coded meaning dimensions forecast Time 2 felt meaningfulness, exhaustion, or turnover intentions; estimates were near zero, and equivalence at the $|\beta| = .30$ sensitivity band held for three of five. One exploratory association survived multiplicity screening: narratives grounding meaning in competence were associated with higher later turnover intentions and exhaustion ($\beta = .29$ and .30, adjusting for six survey measures). Automated coding did not reach the human-agreement standard; humans coded all predictors. Study 2 coded 7,695 nursing-forum posts captured 2016–2026 for meaning events. Nurses named AI in seven events; about one post in twenty per period instead named an optimization or implementation object (ratios, metrics, dashboards, documentation load), and the facet at issue was control over means. Study 3 found that share unchanged across the decade; the one strain marker that rose, lacking control, is confounded with a site template change. No study measured individual AI exposure. Against the automation frontier, the results locate the risk of optimization in the tasks workers name as meaningful, not in AI as such.

Keywords: meaning of work, automation, burnout, turnover intentions, large language models

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Organizations in helping professions increasingly treat artificial intelligence (AI) as a way to shift routine work away from clinicians so they can focus on high-value care. Ambient AI documentation tools are associated with lower documentation burden, note-writing time, and after-hours charting, and their adoption with lower burnout and more undivided attention to patients (Albrecht et al., 2025; Lukac et al., 2025; Olson et al., 2025; Shah et al., 2025; Tierney et al., 2024; You et al., 2025). Health systems adopt ambient documentation, predictive monitoring, and automated care coordination under this banner. They frame the result as restoring the human side of care (Olson et al., 2025; Tierney et al., 2024). The premise is intuitive. It rests on an untested assumption: that managers already know which tasks are routine, which are meaningful, and that the two can be cleanly separated.

Field evidence shows that the routine-meaningful boundary can cut through tasks labeled as efficient targets for automation. When UC Davis Health piloted a wearable biosensor system to automate continuous patient monitoring in an oncology unit, it discontinued the device after roughly a year (Schellmann, 2026). Nurses reported detecting deterioration faster than the algorithm in the sickest patients, using cues the sensors could not capture, and the hospital concluded that the alerts offered no clear advantage over existing monitoring (Schellmann, 2026). Frontline accounts of clinical AI report a similar pattern elsewhere: when algorithmic alerts misfire or cannot explain themselves, nurses carry residual risk and close the judgment gap (Schellmann, 2026). What optimization had classified as automatable monitoring proved to be professional judgment central to nursing identity. The episode is what Siggelkow (2007) called a "talking pig": a single instance establishing that a routine–meaningful boundary can run through

a task automated in the name of efficiency. If that boundary is misdrawn, optimization saves time and removes occasions for meaning. In helping professions, meaning is part of the psychological infrastructure that sustains the workforce rather than a mere perquisite of the job (Bunderson & Thompson, 2009).

Automation may have an iatrogenic effect in helping professions if it is implemented without knowing which tasks carry meaning. Those occupations already face chronic vacancies and attrition. The pipeline depends on intrinsic rewards to offset job costs. Federal projections anticipate national shortages of registered nurses through at least 2038. The same horizon includes shortfalls of roughly 100,000 mental health counselors, 77,000 addiction counselors, and 44,000 psychiatrists (National Center for Health Workforce Analysis, 2025a, 2025b). Exposure to robots and AI is associated with job insecurity and maladaptive workplace behavior (Yam et al., 2023). Insecurity about the automation of occupational content, as distinct from job loss, is associated with burnout and health complaints (Roll et al., 2025). Insecurity from human–AI collaboration spills into work–life outcomes (Wu et al., 2024). Comparing oneself with AI coworkers is one reported burnout path (Ma & Li, 2026). Adoption framed as substitution rather than augmentation is associated with lower willingness to stay among caregivers (Jiang et al., 2026). If the tasks automated to reduce burnout are the tasks through which workers experience meaning, then efficiency optimization can become self-defeating. We name that possibility a task-contingent AI paradox. The proposed mechanism is a diseconomy of scale in meaning: efficiency gains from automating meaning-critical tasks may be offset by losses in the psychological resources that sustain the workforce. The question this article addresses is not whether AI can make helping work faster but which tasks are meaning-critical and what follows when those tasks are automated.

Dominant scholarly and practitioner accounts rarely pose that question; they answer it by assumption. Reviews of AI across helping professions converge on a division of labor. AI absorbs documentation, monitoring, triage, decision support, and administrative coordination. Professionals are then freed for direct care, complex judgment, and compassionate interaction (Balan & Gumpel, 2025; Garkisch & Goldkind, 2024; Guo et al., 2024; Han et al., 2026; Hobensack et al., 2024). Empirical evidence for this narrative concentrates on the clerical side. Automating documentation is associated with lower burden and burnout (Albrecht et al., 2025; Olson et al., 2025; Shah et al., 2025; You et al., 2025). Reviews of AI workplace outcomes give little attention to meaning and purpose (Bankins et al., 2024; Pereira et al., 2023). Conceptual treatments often position automation as a net enabler of meaning when deployment is well governed (Bankins & Formosa, 2023; Smids et al., 2020). The few studies that measure experienced meaningfulness under AI-related job change are recent, largely cross-sectional, and outside flagship industrial-organizational journals (Callari & Puppione, 2025; Gemmano et al., 2026; Liu et al., 2026). This literature rests on a shared assumption: that the routine–meaningful partition of tasks is known, stable, and aligned with what can be automated. That assumption is asserted more often than examined.

Two lines of evidence convert the efficiency story into a theoretical tension. Meaning at work is task-contingent and identity-central. It concentrates in particular activities rather than spreading evenly across a job, a reading that follows from the pathway structure Rosso et al. (2010) describe. Automation effects show the same granularity. Generative AI produces large gains on tasks inside its capability frontier and degrades performance on adjacent tasks just outside it (Dell'Acqua et al., 2026). Several tasks now targeted for automation plausibly carry meaning. Automating monitoring, documentation, and analysis can shrink the task identity and

problem-solving that underpin clinicians' intrinsic motivation (Bienefeld et al., 2025). Diagnostic AI intersects the autonomous judgment that professionals defend as central to responsibility (Faulconbridge et al., 2025; Lebovitz et al., 2022). Expecting one's work to be robotized is associated with lower meaningfulness and more burnout symptoms (Turja et al., 2022). Philosophers of technology describe related "achievement gaps" when machines absorb the accomplishments through which workers earn meaning (Danaher & Nyholm, 2021; Parmer, 2024). The relationship between meaning and well-being is not monotonic. Deeply meaningful, calling-infused work can bind workers to self-sacrifice and exhaustion even as it confers purpose (Bunderson & Thompson, 2009; Cardador & Caza, 2012; Schabram & Maitlis, 2017). The routine–meaningful dichotomy therefore fails on two counts: it mislocates where meaning resides, and it assumes that more meaning is uniformly better. What replaces it is a contingency: the effect of automating a task depends on the task's meaning content and on how densely meaning is already concentrated in the work.

We make that contingency analytically tractable by anchoring in the most influential integrative account of why work becomes meaningful. Rosso et al. (2010) distinguished meaning, the type or content of significance a person attaches to work, from meaningfulness, its amount (see also Pratt & Ashforth, 2003). They synthesized a fragmented literature into four pathways through which meaningful work is created or sustained: Individuation (self-agency), Contribution (other-agency), Self-Connection (self-communion), and Unification (other-communion). These pathways are organized by two superordinate dimensions, a 2 × 2 of Agency–Communion (Bakan, 1966) crossed with Self–Others, which we adopt as the confirmatory level of our framework (Figure 1; facet definitions in Appendix A). The architecture matters for the paradox. Helping work is disproportionately other-oriented and

draws meaning from Contribution and Unification pathways, impact on and connection to patients and community. Those relational tasks are the ones the optimization narrative treats as most valuable. Roadmaps for AI in care now name them as candidates for eventual automation while conceding that relational skills lie outside current system capability (Stade et al., 2024), and conceptual accounts locate the meaning at risk in face-to-face contact and complex judgment (Bankins & Formosa, 2023).

Anchoring in Rosso et al. (2010) also lets us extend a foundational framework to a problem its authors could not have anticipated, and to test conjectures they left open. Rosso and colleagues proposed, but could not examine, that the four pathways might combine additively or interactively. They speculated that "there may even be a curvilinear effect through which too many active pathways to meaning are exhausting and lead to burnout" (p. 116). To our knowledge, that dark-side conjecture has not been tested with longitudinal data. A study built around the 2 × 2 can test those questions directly: whether agentic and communal meaning interact, whether self- and other-oriented meaning forecast different outcomes, and whether the density of activated meaning has a turning point beyond which it predicts exhaustion rather than flourishing (Haans et al., 2016). The contribution is to convert a canonical descriptive framework into falsifiable, outcome-linked propositions about when meaning sustains a workforce and when it consumes it, the kind of extension of prior theory to an emerging problem that Rouse et al. (2025) call for.

We tested the argument in three studies (Figure 2). Study 1 is a two-wave critical-incident panel of helping-profession workers. At Time 1, participants described the meaning of their job and wrote structured critical-incident narratives (Flanagan, 1954) about the episodes they found most meaningful. About one year later, Time 2 measured well-being, work

engagement, prosocial motivation, felt meaningfulness, burnout, and turnover. Each narrative was coded for Rosso's pathways. We then modeled Time 1 codes as predictors of Time 2 outcomes. The result is what we term a meaning-criticality map: which pathways of meaning, if any, forecast later well-being, engagement, and retention. The Study 1 data predate the current automation wave. We do not observe AI removing tasks. We establish which meaning pathways are workforce-sustaining and then read the paradox by placing that map against the independently documented automation frontier. That comparison is weaker than an experiment but stronger than inferring causation from adoption that no individual worker can be shown to have experienced. Studies 2 and 3 add out-of-sample discourse and a year-to-year contrast in that discourse.

Coding thousands of narrative cells by hand is slow. Substituting a single automated coder also removes the between-coder check on which confidence in the labels depends. We therefore tested whether a panel of large language models could take over the coding (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024). Each model applied the same instructions; between-model agreement was to supply reliability evidence, with the prior human dual-coding as the reference standard. None of the six pre-specified prompt configurations met the agency acceptance threshold (see Results), so Study 1 analyses use the human composite codes. Design-based supervised learning (Egami et al., 2023) was specified for machine labels and was not applied once those labels were set aside.

The article treats meaningful work as something optimization can destroy as readily as design can create it. The task-contingent AI paradox specifies when an efficiency intervention can harm workers: when it targets meaning-critical, identity-central tasks. Testing Rosso et al.'s (2010) additive-interactive and curvilinear conjectures converts a descriptive map of meaning

into outcome-linked propositions about workforce sustainability. A two-wave panel then shows which meaning pathways, if any, sustain helping-profession workers over a year. A pre-specified test of automated narrative coding against a human reference retained the human codes when the prompt configurations did not reach the pre-specified agreement thresholds.

Existing accounts theorize how meaning is constructed; few theorize how it is lost, and none, to our knowledge, locate that loss in the collision between an optimization logic and a meaning architecture. The rest of the article develops the framework and four hypothesis blocks, reports the three studies, and returns to the paradox and its limits.

Meaningful Work as Workforce Sustenance

Meaning, Meaningfulness, and Four Pathways

Meaningful work is a working psychological resource with documented downstream consequences rather than a perquisite of employment. Experienced meaningfulness is associated with job satisfaction and performance in cross-sectional data (Gemmano et al., 2026). It transmits effects of technology-related job change onto career sustainability (Liu et al., 2026). It also constitutes a distinct component of subjective well-being at work alongside enthusiasm and contentment (Monnot & Beehr, 2014). Following Rosso et al. (2010), we distinguish the meaning of work, the kind of significance a person attaches to it, from meaningfulness, the amount of positive significance the work holds (Pratt & Ashforth, 2003). Meaning is a qualitative, content question: significance of what kind, arising from which experiences. Meaningfulness is a quantitative question: how much. The distinction matters because the AI paradox concerns content. Automation does not reduce global meaningfulness uniformly; it removes specific tasks through which particular kinds of meaning are realized (Bankins & Formosa, 2023; Montefiore et al., 2026).

Rosso et al. (2010) integrated a fragmented literature into four pathways to meaningful work by crossing two foundational dimensions of human experience. The first is Bakan's (1966) distinction between agency (self-assertion, differentiation, competence, and control) and communion (contact, connection, and unity with others). The second is the direction of the meaning-conferring action, toward the self or toward others. Crossing these yields four pathways: Individuation (self-agency; for example, competence, autonomy, and self-esteem realized through work), Contribution (other-agency; for example, significance, impact, and transcendence of self-interest in service of others), Self-Connection (self-communion; for example, authenticity and identity coherence), and Unification (other-communion; for example, shared values and belonging). In the present project these pathways are operationalized through 13 binary facets, aggregated to the four pathways and then to the 2 × 2 superordinate dimensions using an inclusive (logical-OR) rule. The full mapping is reported in Appendix A (Table A1).

Why the 2 × 2 Is the Confirmatory Level

Our confirmatory tests are specified at the 2 × 2 dimensional level (Agency–Communion × Self–Others) rather than at the level of individual facets. As a measurement matter, fine-grained facets applied to short narratives are individually sparse and noisy, whereas aggregation to pathways and dimensions is more reliable. In our human dual-coding, agreement improves monotonically with aggregation and reaches acceptable levels only at the pathway and dimensional levels (see Method). As a theoretical matter, the 2 × 2 is the level at which Rosso et al.'s (2010) open conjectures are stated, about how agentic and communal, and self- and other-oriented, meaning combine. It is therefore the level at which the framework can be most directly extended and tested. Facet- and pathway-level results are reported as secondary and exploratory.

The Task-Contingent AI Paradox

The paradox follows from two independently documented observations. An automation frontier has already crossed into the daily task structure of helping work. A meaning architecture concentrates significance in specific, identity-central tasks. We document each in turn, then derive the paradox and its diseconomy-of-scale mechanism.

The Automation Frontier in Helping Professions

AI has entered the task structure of every major helping profession as deployed or piloted systems rather than as a speculative future. In medicine, ambient AI scribes draft clinical documentation at health-system scale (Olson et al., 2025; Tierney et al., 2024; You et al., 2025). In nursing, LLMs are entering documentation, clinical decision support, patient education, and workflow management, with efficiency the most common design aim (Han et al., 2026; Hobensack et al., 2024). In mental health care, reviews report structured screening, detection, and decision-support applications (Balan & Gumpel, 2025; Guo et al., 2024), including suicide-risk evaluation (Omar et al., 2024); one roadmap proposes staged integration toward progressively autonomous therapeutic roles (Stade et al., 2024), although World Health Organization guidance summarized in that review favors augmentation over replacement (Guo et al., 2024). Generative AI now also trains counselors through simulated clients and role-play (Jeong et al., 2025; Louie et al., 2026; Maurya, 2024). Social work scholarship maps AI applications across profession, agency, and client functions, though empirical implementation studies remain scarce and ethical concerns remain prominent (Garkisch & Goldkind, 2024; Rodríguez-Martínez et al., 2024; Social Work England, 2025). Algorithmic management, including scheduling, routing, monitoring, and performance scoring, is likewise spreading

beyond gig platforms into standard employment settings, where its prescriptive use remains narrow and uncommon (Keegan & Meijerink, 2025; Parent-Rocheleau & Parker, 2022).

The frontier is jagged. Generative AI capability is task-specific rather than job-level, producing large gains on tasks inside the frontier and degraded performance on adjacent tasks outside it (Dell'Acqua et al., 2026). Effects on service work are mixed rather than uniform (Quan et al., 2025). An optimization logic also governs the frontier: automation and augmentation are interdependent rather than alternatives. Systems marketed as augmentation simultaneously automate, and organizations respond to generative AI coverage by deprioritizing the human skills the technology appears to subsume (Devigili et al., 2026; Raisch & Krakowski, 2021). Substituting machine output for human decisions can impair the experiential learning through which professional judgment develops (Balasubramanian et al., 2022). Even beneficial AI assistance can reduce felt agency and pride when the AI's role is salient (Tong et al., 2021). The tasks that remain contested at the frontier are those in which professional meaning is most concentrated: complex or atypical cases and the assessment of suicide risk. Large language models used in psychiatry have difficulty with complex cases and can underestimate suicide risk (Omar et al., 2024). Evaluations of generative tools in mental-health settings also report drop-offs where contextual reasoning, cultural sensitivity, and high-stakes judgment are required (L. Wang et al., 2025; X. Wang et al., 2025). Professionals engage in deliberate boundary work to protect autonomous judgment and epistemic responsibility from algorithmic encroachment (Faulconbridge et al., 2025). High performers whose identity is invested in skilled judgment resist substitution most strongly (SimanTov-Nachlieli, 2025). Critical-care nurse leaders treated human-centered care, patient autonomy, and accountability in AI-driven decisions as conditions

of integration, not extras (Hassan & El-Ashry, 2024). The frontier has not stopped at the clerical boundary; it is moving through tasks whose meaning content has never been mapped.

Localization: Meaning Is Concentrated in Tasks

Localization is the first property of meaning that generates the paradox. Meaning is realized through specific pathways. It therefore concentrates in specific tasks rather than spreading evenly across a job; this is our inference from the pathway structure Rosso et al. (2010) describe. In helping work, tasks that plausibly carry Contribution and Unification meaning overlap tasks now targeted for automation. Examples include advocating for a patient, coordinating care, being present during a crisis, and exercising judgment when protocols fail (Bienefeld et al., 2025; Schellmann, 2026). The qualifier is deliberate: which tasks carry which meaning is what the meaning-criticality map is designed to establish. The extent of this overlap is a question the studies test rather than a premise we assume. Work-design theory predicts the exposure. Digital technologies act on autonomy, skill use, task identity, and relational contact. Automation decisions are therefore work-redesign decisions whether or not they are recognized as such (Parent-Rocheleau & Parker, 2022; Parker & Grote, 2022). Qualitative evidence from critical care shows that automating monitoring, documentation, and analysis can shrink task identity and problem-solving. Clinicians become supervisors of systems rather than agents of care (Bienefeld et al., 2025). Algorithmic mediation of work is likewise associated with weakened identity and belonging (van Zoonen et al., 2024), and normative analysis places belonging and self-development among the conditions for meaningful work that algorithmic management most threatens (García-Ruiz & Rocchi, 2026).

Non-Monotonicity: The Dark Side of Deeply Meaningful Work

Non-monotonicity is the second property: the relationship between meaning and well-being is not uniformly positive. Deeply meaningful, calling-oriented work is double-edged. It can produce exhaustion, self-sacrifice, and vulnerability to exploitation even as it confers purpose (Bunderson & Thompson, 2009; Cardador & Caza, 2012; Schabram & Maitlis, 2017). Rosso et al. (2010) made the corresponding conjecture explicit: too many simultaneously active pathways to meaning may be exhausting. The conjecture has not, to our knowledge, been tested with longitudinal data. Non-monotonicity supplies a boundary condition and a caution for the paradox. More activated meaning is not uniformly protective. A workforce already dense in meaning-critical tasks may be both most rewarded by those tasks and most vulnerable to their loss.

A Diseconomy of Scale in Meaning

From localization we derive the paradox's sign-reversal. Automating a low-meaning task frees time at little psychological cost. Automating a meaning-critical task removes an occasion for meaning even as it frees the same time. The effect of an efficiency intervention is therefore task-contingent, beneficial or harmful depending on the meaning content of the task it removes. The contingency is already visible in the empirical record. AI adoption was not associated with well-being directly but only through changes in the work conditions employees value (Valtonen et al., 2025). Whether workers respond to AI adoption by crafting toward or defensively away from meaningful tasks depends on whether they appraise the adoption as challenge or hindrance (Cheng et al., 2023; Lin et al., 2024). Aggregated across a workforce, systematically automating meaning-critical tasks yields what we term a diseconomy of scale in meaning. The per-unit efficiency gain is real, but so is a per-unit erosion of the psychological resources that sustain

retention and engagement, and the latter accumulates as the former scales. From non-monotonicity we draw the complementary caution. Because meaning can also exhaust, the map of what to protect must be established empirically rather than assumed. Some dense concentrations of meaning may mark burnout risk rather than assets to preserve.

We treat the paradox as a bridging proposition rather than a directly tested hypothesis because Study 1 does not measure AI exposure (see Overview of the Research Program). The falsifiable content of Study 1 is the meaning-criticality map: which pathways predict which outcomes over time. That map is summarized in the hypotheses below and depicted in Figure 1. Studies 2 and 3 then ask whether the same pathways appear in organic discourse and whether their salience moved across the years we could observe.

Hypotheses

We organize predictions into four blocks that move from aggregate to dimensional to pathway-specific effects and, finally, to the curvilinear dark side. Blocks 1-3 concern linear, workforce-sustaining relationships. Block 4 tests Rosso et al.'s (2010) curvilinear conjecture. A compact summary appears in Appendix *B* (Table B1). Throughout, predictors are Time 1 (T1) meaning coded from critical-incident narratives, and outcomes are Time 2 (T2) measures collected approximately one year later.

Block 1: Aggregate Main Effects

The aggregate proposition that meaning sustains a workforce has converging empirical footing. Experienced meaningfulness is associated with satisfaction and performance (Gemmano et al., 2026). It carries the effects of technology-related job change onto career sustainability (Liu et al., 2026). Expecting one's work to be robotized is associated with lower meaningfulness and more burnout symptoms (Turja et al., 2022), and substitution-framed adoption with reduced

willingness to stay (Jiang et al., 2026). What this literature lacks is prospective evidence that the meaning content expressed in workers' own accounts of their work predicts consequential outcomes over time. Block 1 supplies that test at the aggregate level, treating the breadth of activated pathways as the coarsest index of a worker's meaning architecture. Block 1 predictions are stated as first-order, average linear relationships. Whether the linear form holds across the observed range is itself tested in Block 4.

Hypothesis 1a. The overall density of activated T1 meaning (the breadth of pathways present in a worker's meaningful-work narrative) positively predicts T2 affective well-being and work engagement.

Hypothesis 1b. Aggregate T1 meaning density positively predicts T2 prosocial motivation.

Hypothesis 1c. Aggregate T1 meaning density negatively predicts T2 turnover intentions and voluntary mobility.

Hypothesis 1d. Aggregate T1 meaning density positively predicts T2 felt meaningfulness of work. This hypothesis establishes the predictive bridge between coded meaning content and experienced meaningfulness as distinct constructs (Pratt & Ashforth, 2003), using a self-report meaningfulness outcome (Monnot & Beehr, 2014) that shares no method variance with the narrative-coded predictor.

Block 2: Dimensional (2 × 2) Confirmatory Effects

Helping work is disproportionately other-oriented. Its meaning is anchored in impact on, and connection to, beneficiaries (Grant, 2008b; Rosso et al., 2010). Helping professionals themselves locate the non-negotiable core of their occupations in the relational and moral dimensions of care (Bienefeld et al., 2025; Hassan & El-Ashry, 2024). The burnout prediction in

this block follows from construct correspondence. Depersonalization is the interpersonal component of burnout, a cynical withdrawal from the relational field of work, whereas emotional exhaustion is its general energetic component (Maslach et al., 2001). Meaning anchored in others should therefore most strongly buffer the burnout component that shares its relational content. Block 2 specifies confirmatory tests on the contrast-coded Agency-Communion and Self-Others dimensions and their interaction. These tests distinguish Rosso et al.'s (2010) additive account from an interactive one.

Hypothesis 2a. The Self–Others contrast predicts T2 workforce-sustaining outcomes such that other-oriented T1 meaning is more strongly and positively associated with T2 well-being and engagement than self-oriented meaning.

Hypothesis 2b. Agency and Communion interact rather than combine purely additively in predicting T2 outcomes; the effect of one dimension depends on the level of the other (a non-additive 2 × 2 interaction).

Hypothesis 2c. The Self–Others contrast differentially predicts burnout components: other-oriented T1 meaning is more strongly (inversely) associated with T2 depersonalization than with T2 emotional exhaustion; this difference warrants separate treatment of these subscales in this block.

Block 3: Pathway-Specific Linear Effects

Within the other-oriented dimension, the two pathways rest on theoretically distinct mechanisms. Contribution operates through perceived impact and significance, the mechanism by which contact with beneficiaries fuels prosocial motivation and persistence (Grant, 2008a, 2008b). Unification operates through shared values and belonging, the relational resources most directly threatened when algorithmic mediation weakens identity and connection at work (van

Zoonen et al., 2024). Outcome assignments follow these mechanisms. Contribution's impact mechanism feeds motivational and affective outcomes. Unification's belonging mechanism is an embedding force: shared values and connection bind workers to the setting that supplies them. Unification, uniquely among the pathways, therefore carries the turnover and mobility prediction. Block 3 tests whether these pathways carry the workforce-sustaining effects that the paradox would place at risk.

Hypothesis 3a. The T1 Contribution pathway positively predicts T2 well-being, engagement, and prosocial motivation.

Hypothesis 3b. The T1 Unification pathway positively predicts T2 engagement and prosocial motivation and negatively predicts T2 turnover and mobility.

Hypothesis 3c. The self-oriented pathways (Individuation, Self-Connection) predict workforce-sustaining outcomes less consistently than the other-oriented pathways (Contribution, Unification).

Block 4: Curvilinear, Dark-Side Effects

Block 4 tests the conjecture that "too many active pathways to meaning are exhausting and lead to burnout" (Rosso et al., 2010, p. 116), using the theorizing-and-testing guidelines for U-shaped relationships set out by Haans et al. (2016). Block 4 stands in a specified relation to Block 1 rather than alongside it: Hypotheses 1a and 4c address the same predictor–outcome pair with competing functional forms, and Hypotheses 1a and 4a do so for near-mirror outcomes. We therefore treat Block 1 as stating the average, first-order relationship and Block 4 as testing whether that relationship is genuinely linear or conceals a turning point, with the quadratic specification as the arbiter between them (Haans et al., 2016); a quadratic term estimated with an interval excluding zero would refine rather than complement the aggregate proposition. That

places its validity in the pre-turning-point range. The dark-side literature locates the exhaustion risk specifically in other-oriented, calling-infused meaning, where self-sacrifice and over-identification bind workers to the source of their depletion (Bunderson & Thompson, 2009; Cardador & Caza, 2012; Schabram & Maitlis, 2017).

Hypothesis 4a. The density of activated T1 meaning pathways has a curvilinear relationship with T2 burnout (emotional exhaustion): beyond a turning point, additional activated meaning predicts higher, not lower, exhaustion.

Hypothesis 4b. The curvilinear density–burnout relationship is stronger for other-oriented (Contribution, Unification) meaning than for self-oriented meaning, consistent with the dark side of calling-based, service-oriented work.

Hypothesis 4c. The density of activated T1 meaning has an inverted-U relationship with T2 affective well-being: moderate density predicts the highest well-being, whereas very high density predicts diminishing and ultimately declining well-being.

Overview of the Research Program

The argument is tested in three completed studies (Figure 2). Study 1 is the two-wave panel. Study 2 coded a decade of nursing-forum posts for meaning events and the objects they implicate. Study 3 tested whether that talk shifted across the 2016–2026 capture window.

Study 1 identified meaning-critical tasks in a two-wave critical-incident panel of helping-profession workers. Time 1 meaning pathways were used to predict Time 2 well-being, engagement, prosocial motivation, felt meaningfulness, burnout, and turnover.

Study 2 applied the Study 1 facet scheme, extended to record polarity and the object implicated, to organic occupational discourse from a nursing forum. The questions were whether

the same meaning pathways appear outside a survey frame and what, if anything, nurses name as the source of frustrated meaning.

Study 3 tested whether that talk changed across capture years 2016–2026, including the years after generative AI became widely available. Capture year is a proxy for posting year, so the test is a contrast among archive windows, not a measured adoption effect.

Study 1 measured meaning and outcomes, not AI exposure. The paradox is therefore read by placing the meaning-criticality map against the independently documented automation frontier, not by a measured AI effect. Study 2 tested whether the pathways appear in organic discourse and what objects nurses attach to them. Study 3 tested whether their prevalence moved across the available years. Neither study measured individual-level AI use.

Study 1 Method

Participants and Procedure

Helping-profession workers were recruited through Amazon Mechanical Turk and completed measures at two waves about one year apart. Time 1 was collected March 10–16, 2016. Participants described the meaning they associate with their job, wrote structured critical-incident narratives (Flanagan, 1954) about the work episodes they found most meaningful, and completed the Time 1 scales below. Time 2 was fielded February 14 to March 30, 2017, under an IRB modification signed December 27, 2016. Across matched respondents, the interval between wave start dates ranged from 11.1 to 12.6 months (M = 11.58, SD = 0.28). The Time 1 narratives predate public release of contemporary generative-AI systems. The analytic sample is 307 respondents at Time 1 and 144 longitudinal survey cases at Time 2 (pre-specified models n = 136 and 128). The sample is majority health care (roughly 55%). Time 2 completers were older than

noncompleters (M = 38.1 vs. 33.7 years, d = 0.43, $p < .001$), higher in work centrality (d = 0.25, p = .029), and slightly lower in Time 1 meaningfulness (WAMI $d = -0.22$, p = .062).

The critical-incident technique fits the localization premise. Rather than asking workers only to rate global meaningfulness, it elicits concrete episodes. Meaning can then be coded at the level of specific tasks and events (Flanagan, 1954). Each narrative was collected in three parts (context, event, and outcome). That three-part record is the unit of coding.

Time 1 Measures

Job-meaning description. An open-ended prompt asked participants to describe the meaning they associate with their job. This global account was coded separately from the critical-incident narratives.

Critical-incident narratives. Participants wrote three-part narratives (context, event, outcome) of their most meaningful work experiences (Flanagan, 1954). These narratives are the source text for the meaning codes.

Meaningful work (WAMI). The 10-item Work and Meaning Inventory (Steger et al., 2012) was administered at Time 1 and again at Time 2. A sample item is "I have found a meaningful career." The reverse-keyed item is "My work really makes no difference to the world." Respondents used a 5-point scale (1 = absolutely untrue to 5 = absolutely true). The Time 1 total served as the self-report meaningfulness covariate in prospective models (α = .92). The Time 2 total was a secondary meaningfulness outcome (α = .93).

Meaningfulness of work (SWB-M). The six-item meaningfulness component of job-related subjective well-being (Monnot & Beehr, 2014) was also administered at both waves. Reverse-keyed items asked whether work felt "irrelevant" or "trivial." This scale is the Hypothesis 1d outcome at Time 2. It shares no method with the narrative codes.

Work centrality. A single item assessed how central work was to the respondent's life. Completers differed from noncompleters on this item, so it is a covariate in the longitudinal models.

Time 2 Measures

Time 2 outcomes spanned affective, motivational, and behavioral consequences relevant to staying in the work. Unless noted, composites are item means. A construct-by-wave list is in Appendix C (Table C1).

Work engagement. State work engagement was measured with the nine-item Utrecht Work Engagement Scale adapted for state assessment (Breevaart et al., 2012). A sample item is "At my work, I feel bursting with energy." Respondents used the standard UWES frequency anchors (0 = never to 6 = always; $\alpha = .94$).

Prosocial motivation. Items from Grant (2008a) assessed motivation to benefit others through work. A sample item is "I care about benefiting others through my work." Grant's published scale uses a 7-point format (1 = disagree strongly to 7 = agree strongly). The Time 2 composite used here had $\alpha = .91$.

Affective well-being. Job-related affect was measured across activation and pleasantness quadrants (Warr et al., 2014). Respondents reported how much of the time their job made them feel states such as enthusiastic and gloomy (frequency ratings from never to all of the time).

Meaningfulness of work. Time 2 meaningfulness was the SWB-M meaning component (Monnot & Beehr, 2014) and the WAMI total (Steger et al., 2012), as described under Time 1. Both are self-report outcomes distinct from the narrative-coded predictor (Hypothesis 1d).

Burnout. Emotional exhaustion and depersonalization were taken from the Maslach Burnout Inventory (Maslach et al., 2001). A sample exhaustion item is "I feel emotionally

drained from my work." A sample depersonalization item is "I feel I treat some recipients as if they were impersonal objects." The published frequency anchors are 0 = never to 6 = every day. Exhaustion and depersonalization were analyzed separately; only exhaustion entered the pre-specified tests, and depersonalization appears in the exploratory screen.

Turnover and mobility. Turnover intention was an eight-item composite of intention to leave and voluntary mobility indicators (no Time 1 administration; α not separately tabled here). Higher scores indicate stronger exit cognition.

Table C1 lists constructs, wave of administration, and source instruments.

Coding Meaning From Narratives

The central measurement task was to code each critical-incident narrative for the meaning pathways it expressed. This subsection covers the coding framework, the human dual-coding that produced the Study 1 predictors, the agreement metric, and the pre-specified machine-coding test that did not support substitution.

Coding Framework

Narratives were coded on 13 binary (present/absent) facets adapted from Rosso et al. (2010) and specified in the project coding instructions. Facets were aggregated to the four pathways and then to the 2 × 2 dimensions with an inclusive logical-OR rule: a pathway is present if any of its facets is present, and a dimension is present if any of its pathways is present. Appendix A (Table A1) reports the mapping. We treat the four pathways and the 2 × 2 dimensions as the valid aggregation layers. The seven narrative mechanisms in Rosso et al. (2010) are not a coding level in this scheme.

Human Coding as the Validation Reference

Two coders independently applied the codebook to the Time 1 narratives: the first author and a trained research assistant. Disagreements on a stratified sample of cells were then resolved by the first author, blind to which coder had assigned each label (identity keys stayed sealed until that pass was finished). This is a blinded resolution pass by one of the original coders, not an independent third-rater design. Between-coder agreement was computed on the two original independent passes and is unaffected by the later resolved codes. The composite (agreed cells plus resolved cells) is the human reference used in Study 1. It is not a machine-learning training set. Because joint absence exceeds roughly 95% of cells, reliability is reported as the proportion of specific agreement (PSA; Cicchetti & Feinstein, 1990). PSA equals the F1 score when one coder is treated as the reference and the other as the prediction (van Rijsbergen, 1979; see also Dice, 1945). Unlike kappa, PSA is not deflated by extreme class imbalance (Feinstein & Cicchetti, 1990). Agreement was lowest at the facet level and higher at the pathway and dimensional levels. That pattern is one reason the confirmatory tests are specified at the 2 × 2 level. Cells still unresolved after the resolution pass (9.3%) were coded absent in the primary analysis. An endpoint sensitivity check recoded those cells as present. The two-tier coding process is summarized in Figure 3.

The LLM Panel as Candidate Coders

To test whether coding could be scaled, we treated a panel of large language models as candidate coders. Each model applied the same written instructions and returned a present/absent label per facet. Reliability was estimated as between-model PSA/F1, the same index used for the human pair (Cicchetti & Feinstein, 1990; van Rijsbergen, 1979). The human reference was used to test whether panel labels matched competent human judgment (Gilardi et al., 2023; Törnberg,

2024; Ziems et al., 2024). Substitution was conditional on meeting pre-specified agreement thresholds on the development partition, not on a single validation.

The primary machine-coding arm was a few-shot panel. A small set of human-resolved exemplars anchored each facet definition in the prompt. Zero-shot and one-shot prompting were baseline and ablation conditions. Development and evaluation cases did not overlap. Prompts and examples were labeled by provenance so that no fabricated text was presented as a real narrative and no illustrative label was presented as a resolved human code. Agreement was reported as PSA/F1 and confusion structure (Cicchetti & Feinstein, 1990; van Rijsbergen, 1979), not as a correlation, because the coding target is categorical.

Fine-tuning is discussed only as a conceptual benchmark. Customer fine-tuning is not currently available for the frontier-tier model used here, so a fine-tuned arm was not executable by an external research team and was not run. We mention it to bound expectations about achievable accuracy.

Automated Coding: Pre-Specified Validation Against the Human Benchmark

We tested whether large language model (LLM) coding could substitute for human dual-coding before any substantive analysis. Six prompt configurations were developed exclusively on the training partition and fixed by SHA-256 hash before evaluation: a decision-first instruction and a revised decision-first instruction, a per-facet version that issued thirteen separate calls per narrative, a reason-first variant, a decision-plus-evidence variant, and an alternate-codebook variant. Each prompt was evaluated once on the 55-case development partition under a selection rule fixed in advance: three replicates per prompt on the pre-specified primary model (claude-sonnet-5, run 2026-09-22), selection statistics computed on counts pooled across replicates, and case-level cluster bootstrap (B = 10,000, fixed seed) for all intervals. A

second model (gpt-5-2025-08-07) ran a single non-voting pass used only for salience reliability. Two development-set exposures predated the freeze: an earlier model generation had scored 28 of the 55 cases in July 2026, and the choice of communion precision as a development target traced partly to that exposure. The pass was therefore recorded as the single selection evaluation, with this exposure disclosed, rather than treated as a first evaluation on unexposed data.

Acceptance was calibrated to what human coders achieve. A prompt met the standard set for acceptance only if its dimension-level agreement with the composite gold standard (positive specific agreement; PSA) reached the lower bound of the human–human benchmark's bootstrap 95% confidence interval: ≥ .910 for agency (human PSA = .935 [.910, .956]) and ≥ .805 for others-directed meaning (human .844 [.805, .879]). Communion (human .48 [.34, .60]) and self-directed meaning (human .52 [.43, .59]) carried no acceptance threshold because human agreement was itself too low to validate against. Disqualifiers, evaluated over pooled replicate cells, were any false positive on communion facets or on sparse facets over cells both human coders had coded absent, and a structured-output error rate above .10.

None of the six prompt configurations met the agency threshold. Agency PSA was .750–.798 against ≥ .910; the highest value, .798, fell .11 below the threshold. Two decision-first prompts met the others-directed threshold (≥ .805). Full prompt-level numbers are in Table S1.

Neither provider's interface accepted a fixed temperature value for these models, so each provider's default was used for every arm. Default parameters must be specified even when they are defaults, for reproducibility (Törnberg, 2024); we record them here as the providers' defaults. That constraint was one reason for the replicate design. An interruption in the middle of the run, and the resume that followed, meant that a small number of initially malformed outputs were redrawn, so error rates are quantities from the final attempt. Interim aggregate monitoring

occurred during the run with no subsequent change to any prompt, freeze, or rule. No model scoring of the test partition occurred before the results above were fixed. After the development pass and after these analyses were complete, one descriptive evaluation of a pre-specified 65-case subsplit of the test partition was taken under a hashed protocol that fixed the arms, statistics, and reporting rule before any case was scored; it scored the two highest-ranked disqualified prompts. On those 65 cases agency PSA was .81 (decision-first) and .80 (per-facet), and others-directed PSA .75 and .77, in the range of the development pass; both prompts again carried false positives over agreed-absent cells. The pre-specified screen for scoring the remaining 128 cases (decision-first agency and others-directed PSA both ≥ .75) was not met (others-directed = .746), so those cases remained sealed under that screen. A single, separately authorized scoring pass over the 128 was underway at the time of writing and is not reported here. Table S2 reports the estimates with same-case human-pair comparators and a leakage sensitivity; they are descriptive, at development-level precision, and not a validation of the instrument.

Analytic Strategy

Confirmatory 2 × 2 Model

The confirmatory analyses regressed each T2 outcome on contrast-coded Agency–Communion and Self–Others dimensions and their interaction, rather than on four collinear pathway dummies. Contrast coding yielded interpretable main effects of the two superordinate dimensions and a single interaction term that directly tested the additive-versus-interactive conjecture (Block 2). Pathway-specific (Block 3) and facet-level analyses were estimated as secondary and exploratory models.

Estimation, Covariates, and Inference Criteria

Each T2 outcome was modeled with ordinary least squares, with the hypothesized predictors entered simultaneously; no stepwise or best-subset selection was used. Every model carried a fixed covariate set: the T1 baseline of the outcome where one exists (the T1 meaningfulness measure in the felt-meaningfulness model, which makes Hypothesis 1d a test of incremental prediction over self-reported meaning), age and work centrality (the variables on which completers differed from noncompleters), and occupation block. Coefficients are reported unstandardized (Table 2) and standardized (Table 3) with heteroskedasticity-consistent (HC3) standard errors (Long & Ervin, 2000) and 95% confidence intervals, with the complete-case n for each model. Pre-specified directional hypotheses were tested at $\alpha = .05$, one-tailed in the predicted direction; the non-directional interaction (Hypothesis 2b) and comparisons of functional form (Block 4 against Block 1) were two-tailed. The confirmatory family was deliberately small. All remaining dimension-, pathway-, and facet-level estimates are reported in full as point estimates with confidence intervals and are not treated as hypothesis tests, so no multiplicity adjustment was applied to associations that were never tested. Analyses were run in Python (pandas, statsmodels, SciPy).

Curvilinear Tests

Curvilinear predictions (Block 4) were to be tested following Haans et al. (2016): a quadratic term for meaning density, the significance and sign of the quadratic coefficient, a turning point within the observed data range, and a slope that reverses across that range rather than merely flattening. Where a dark-side effect was predicted to differ by orientation (Hypothesis 4b), curvature was to be compared across other- and self-oriented meaning. Because Block 4 respecifies the functional form of relationships stated as linear in Block 1, direct

comparison of the linear and quadratic specifications decides between them; where the quadratic term's interval excludes zero, Block 1 coefficients are interpreted as average effects over the observed range.

Statistical Power and What the Sample Could Test

With 144 complete longitudinal cases, the design detects zero-order associations of $|r| \geq .23$ with 80% power ($\alpha = .05$, two-tailed); effects that must additionally clear an autoregressive baseline need to be larger to be detected. Two consequences were fixed in advance rather than decided after inspection. First, confirmatory tests were specified at the 2 × 2 dimensional level, where human-coded base rates supply usable variance: Agency and Others are present in roughly one half to two thirds of human-coded narratives, whereas the Self-Connection and Unification pathways each appear in fewer than 10%. Pathway-level predictions that depend on those sparse pathways (Hypotheses 3b and 3c) were therefore evaluated as estimation with confidence intervals rather than as formal tests. Second, a hypothesis was withheld from formal testing and reported descriptively when the data could not satisfy the design requirements of its test. Both pre-set conditions were met: only 16 respondents carried both Agency and Communion, so the interaction (Hypothesis 2b) was not estimable, and the density distribution concentrated at 0–2 activated pathways, leaving no interior turning point for the Haans et al. (2016) criteria to certify (Block 4). Declared non-tests are reported with the relevant cell counts and distributions and are not interpreted as null results.

Debiasing Downstream Inference

Any coder, human or machine, measures the Time 1 meaning predictor with error. Naive regression of Time 2 outcomes on imperfect labels yields biased coefficients (Egami et al., 2023). Design-based supervised learning (DSL; Egami et al., 2023) was specified for that

problem if machine labels had been used. The pre-specified machine-coding test did not support substitution, so DSL was not applied. The Study 1 models use the human composite codes.

Missing Data and Attrition

Because Time 2 was available for a subset of Time 1 respondents, we compared completers and noncompleters on Time 1 variables (reported above) and entered age and work centrality, the variables on which they differed, as covariates in every model. Longitudinal models use complete cases and report their *n*; Table 1 correlations use pairwise deletion. No imputation was performed.

Study 1 Results

Pre-Specified Predictions

Study 1 asked whether Time 1 coded meaning dimensions forecast Time 2 felt meaningfulness, emotional exhaustion, or turnover intentions. Descriptive statistics and intercorrelations among the study variables appear in Table 1. Predictors were the human composite codes: person-level presence of each dimension, formed by disjunction over facet codes. Models were ordinary least squares with HC3 standard errors (Long & Ervin, 2000), with age, work centrality, occupation block, and the Time 1 baseline of the outcome when the same construct was measured at both waves. Directional tests were one-tailed at $\alpha = .05$. Confidence intervals are two-sided.

Table 1

Descriptive Statistics and Intercorrelations Among Study Variables

Variable	***M***	***SD***	1	2	3	4	5	6	7	8	9	10	11
1. Agency (code present)	0.64	0.48	—										
2. Others-directed (code present)	0.56	0.50	.71	—									
3. Self-directed (code present)	0.23	0.42	.30	−.16	—								
4. Communion (code present)	0.11	0.32	−.09	.13	.22	—							
5. Meaningfulness, T1 (WAMI)	3.97	0.78	.20	.24	.10	.13	—						
6. Meaning of work, T1 (SWB-M)	4.44	1.05	.11	.16	.06	.12	.80	—					
7. Work centrality, T1	2.52	1.45	−.14	−.09	−.13	−.08	−.36	−.30	—				
8. Meaningfulness, T2 (WAMI)	3.86	0.79	.04	.16	−.05	.21	.57	.48	−.18	—			
9. Meaning of work, T2 (SWB-M)	4.35	1.09	.07	.14	−.05	.20	.46	.53	−.26	.77	—		
10. Turnover intentions, T2	2.34	1.35	.05	.00	.15	−.14	−.13	−.26	.10	−.32	−.46	—	
11. Emotional exhaustion, T2	2.50	1.17	−.01	−.05	.17	.01	−.20	−.27	.01	−.33	−.43	.46	—

Note. Time 1 $n = 272$ coded cases with survey linkage; Time 2 $n = 132–140$. Variables 1–4 are binary presence indicators from the adjudicated composite gold coding of Time 1 narratives; correlations involving them are point-biserial. Meaningfulness = Work and Meaning Inventory;

SWB-M = subjective well-being meaning-of-work item set. Pairwise deletion. Correlations ≥ |.17| are significant at $p < .05$ for the smallest pairwise n.

Table 2 and Figure 4 report the six pre-specified tests (five coefficients and one block test). Others-directed meaning did not predict Time 2 felt meaningfulness ($b = -0.02$, 95% CI [−0.53, 0.49], one-tailed $p = .537$, $n = 136$). Its association with turnover intentions ran opposite to prediction ($b = 0.22$ [−0.65, 1.09], $p = .688$). Agency was unrelated to emotional exhaustion ($b = -0.29$ [−1.20, 0.62], $p = .268$, $n = 128$) and to felt meaningfulness ($b = 0.05$ [−0.47, 0.57], $p = .428$). Self-directed meaning, predicted to relate positively to felt meaningfulness, trended negative ($b = -0.31$ [−0.73, 0.12], $p = .921$). The three-dimension block added $\Delta R^2 = .012$ over Time 1 self-reported meaning and covariates, robust Wald $F(3, 126) = 0.73$, $p = .534$ (full-model $R^2 = .35$). No interval excluded zero in the predicted direction.

The high-confidence ($n = 67$) and flag-excluded ($n = 101$) re-estimates still had intervals covering zero. Recoding the 9.3% of unresolved facet cells as present left the block at $\Delta R^2 = .003$; individual upper-bound coefficients moved (Others → turnover $b = -0.35$; Agency → meaningfulness $b = 0.26$) and their intervals still included zero. Restricting each model to persons whose focal dimension was determinate from resolved cells alone did the same.

Communion was excluded from the confirmatory family before estimation because human–human agreement was too low (PSA = .48, 95% CI [.34, .60]). The unadjusted associations with Time 2 felt meaningfulness ($b = 0.40$ [0.02, 0.78]) and turnover intentions ($b = -0.97$ [−1.66, −0.27]; $n = 136$) were not robust to unresolved cells. At the present-coded bound the meaningfulness estimate fell to $b = 0.04$ [−0.36, 0.45] and the turnover interval crossed zero.

Two pre-specified questions could not be tested. Only 4 of 276 persons narrated three or more pathways, so the curvilinear density tests had no interior turning point. Only 16 persons carried both agency and communion, so the interaction was not estimable.

Table 2

Confirmatory Tests of the Pre-Specified Predictions

Prediction	Predictor and T2 outcome	Dir.	***b***	95% CI	***p***	***n***
P1	Others → felt meaningfulness	+	−0.02	[−0.53, 0.49]	.537	136
P2	Others → turnover intentions	−	0.22	[−0.65, 1.09]	.688	136
P3	Agency → emotional exhaustion	−	−0.29	[−1.20, 0.62]	.268	128
P4	Agency → felt meaningfulness	+	0.05	[−0.47, 0.57]	.428	136
P5	Self → felt meaningfulness	+	−0.31	[−0.73, 0.12]	.921	136
P6	Joint block over T1 meaning	$\Delta R^2 > 0$	$\Delta R^2 = .012$	—	.534 (two-tailed)	136

Note. OLS with HC3 standard errors; covariates: age, work centrality, occupation block, plus the T1 baseline for the meaningfulness models. Confidence intervals are two-sided. P6 is the robust

Wald test of the three-dimension block over Time 1 self-reported meaning and covariates. Dir. = hypothesized sign. Tests are one-tailed except P6.

Equivalence tests (Lakens, 2017) and BIC-approximated Bayes factors (Wagenmakers, 2007) bound these estimates and separate small effects from low power (Table 3). These supplements were added after the primary tests; their interpretation criteria were fixed before the numbers were computed. For the three-dimension block over Time 1 self-reported meaning, BF_{01} = 481.5. Per-dimension Bayes factors ranged from 4.7 to 11.6. At the sensitivity band |β| = .30, equivalence held for three of five predictions in the simultaneous specification (P1, P4, and P5); at the primary band |β| = .20, it held for none of five. Agency and Others co-occur strongly (φ = .71), which widens the intervals. Entered alone, equivalence at |.20| held for four predictions (p = .006–.023) and at |.30| for all five.

Table 3

Equivalence Tests and Bayes Factors for the Pre-Specified Prospective Predictions

Prediction	*β*	90% CI	TOST *p* (±.20)	TOST *p* (±.30)	BF_{01}	*n*
P1 Others → Meaning of work	−.011	[−.205, .183]	.055	.008	11.6	136
P2 Others → Turnover intentions	.079	[−.188, .345]	.226	.086	9.9	136
P3 Agency → Emotional exhaustion	−.116	[−.427, .195]	.328	.164	8.2	128
P4 Agency → Meaning of work	.021	[−.169, .211]	.061	.008	11.5	136
P5 Self → Meaning of work	−.119	[−.259, .021]	.171	.017	4.7	136
P6 Three-dimension block over T1 meaning	—	—	—	—	481.5	136

Note. Post hoc supplement to the pre-specified tests; interpretation criteria stated before computation. β = standardized coefficient from the pre-specified simultaneous specification (three dimensions entered together; age, work centrality, occupation block, and Time 1 baseline covaried; HC3 standard errors). TOST = two one-sided tests against smallest effect sizes of

interest |β| = .20 (primary) and .30 (sensitivity). BF_{01} = BIC-approximated Bayes factor (Wagenmakers, 2007) favoring the null over the focal-predictor alternative.

Associations That Survived Screening

Four associations survived Benjamini–Hochberg screening at $q = .10$ (Benjamini & Hochberg, 1995) in a disclosed search of about 1,700 tests. They are exploratory. In descending order of importance: (a) Time 1 competence-anchored meaning was associated with higher Time 2 turnover intentions ($b = 1.40$, 95% CI [0.58, 2.22], $p < .001$, $n = 100$) and higher Time 2 emotional exhaustion ($b = 0.95$ [0.20, 1.70], $p = .014$, $n = 94$); (*b*) coded dimensions converged with concurrent WAMI scores (others-directed $r = .24$, $p_{BH} = .002$; agency $r = .20$, $p_{BH} = .011$; contribution $r = .23$, $p_{BH} = .005$; dimension-block $\Delta R^2 = .08$); (c) meaning claimed in the abstract answer but absent from the story was associated with concurrent WAMI and with Time 2 turnover intentions ($b = 0.80$ [0.27, 1.33], $p = .003$); (d) coder disagreement was higher when concurrent meaningfulness was lower ($b = -0.21$, $p = .004$, $n = 272$). No moderation, retention, job-satisfaction, affective well-being, or work–life association survived screening.

Competence-Anchored Meaning and Later Strain

Participants whose Time 1 narratives grounded meaning in competence or mastery reported higher turnover intentions one year later ($b = 1.40$, 95% CI [0.58, 2.22], $p < .001$, $n = 100$, adjusting for age, work centrality, and occupation). They also reported higher Time 2 emotional exhaustion ($b = 0.95$ [0.20, 1.70], $p = .014$, $n = 94$). Neither outcome was measured at Time 1, so incremental validity over a baseline could not be tested. Figure 5 shows the unadjusted cell means. Treating turnover ordinally gave an adjusted ordered-logit odds ratio of 5.47 [1.94, 15.42] (rank-biserial $r = .42$). The coefficient was sign-consistent in all 200 random split-half re-estimations. Two constraints remain. The facet's base rate is low (17 of 140 matched

cases; 12 coded present by both coders, 5 resolved in the disagreement pass). The association holds under the composite codes and under the second coder's sparser codes (r = .26), but not under the first coder's unfiltered codes (r = .02). The code records that competence appeared as a meaning source. It does not distinguish satisfied competence from frustrated competence, contingent self-worth, or perceived mobility.

A benchmark analysis conducted after these results sharpens their interpretation. The WAMI total, the positive-meaning and meaning-making subscales, and the SWB-M items forecast Time 2 meaning at r = .43–.54 (same-construct stabilities r = .51 and .52 in the full panel, n = 144; .57 and .53 among the coded cases in Table 1, n = 136). Greater-good motivation forecast those outcomes at r = .19, and work centrality at r = −.22 to −.27. The same battery forecast strain and exit weakly (strongest single measure: SWB-M, r = −.26 with turnover intentions and r = −.28 with emotional exhaustion). Against that benchmark the binary competence code was the strongest single prospective correlate on the common complete-case n (turnover r = .29, n = 100; exhaustion r = .27, n = 94; available-case correlations r = .28, n = 140 and 132), and it retained its association with all six survey measures and age entered jointly (turnover: β = .29, 95% CI [.04, .54], p = .021; exhaustion: β = .30, 95% CI [.06, .55], p = .015; Table 4). The earlier caution that no Time 1 baseline exists for these outcomes therefore understates the case: the code adds prospective signal beyond the strongest quantitative control set this design affords. The estimate remains exploratory and rests on 17 positive cases; nothing about the benchmark changes its status as a candidate effect awaiting a designed test.

Table 4

Robustness of the Exploratory Association Between Competence-Coded Narratives (Time 1) and Time 2 Strain and Exit Cognitions

Model	Time 2 outcome	Estimate	95% CI	***p***	***n***
OLS, unadjusted	Turnover intentions	$r = .28$	[.12, .43]	< .001	140
OLS, adjusted[a]	Turnover intentions	$b = 1.40$	[0.58, 2.22]	< .001	100
Ordered logit, adjusted[a]	Turnover intentions (5 bins)	OR = 5.47	[1.94, 15.42]	.001	136
Mann–Whitney	Turnover intentions	$r_{rb} = .42$	—	.005	140
OLS, unadjusted	Emotional exhaustion	$r = .28$	[.11, .43]	.001	132
OLS, adjusted[a]	Emotional exhaustion	$b = 0.95$	[0.20, 1.70]	.014	94
OLS, full T1 survey battery + age[b]	Turnover intentions	$\beta = .29$	[.04, .54]	.021	100
OLS, full T1 survey battery + age[b]	Emotional exhaustion	$\beta = .30$	[.06, .55]	.015	94

Note. Exploratory, post hoc; part of a disclosed ~1,700-test search (Benjamini–Hochberg $q = .10$ within family). Competence code present for 17 of 140 matched cases (12 both-coder, 5 adjudicated). [a] Adjusted for age, work centrality, and occupation block; OLS models use HC3 standard errors. Row n differ by coding rule: the adjusted OLS rows use cases whose competence cell was resolved by both coders or by adjudication ($n = 100$ turnover, 94 exhaustion); the unadjusted and ordered-logit rows code unresolved competence cells as absent ($n = 140$, 132, and 136). [b] Adjusted for all six Time 1 survey measures (WAMI total, SWB-M, the three WAMI subscales, work centrality) plus age, entered jointly; β = standardized

coefficient. Neither outcome was measured at Time 1, so same-construct baseline adjustment is not possible. r_{rb} = rank-biserial correlation. Split-half re-estimation: 100% sign-consistent across 200 random halves (turnover 61% and exhaustion 39% also significant per half).

Concurrent Convergence With Self-Reports

Although the coded dimensions added no prospective information over Time 1 self-reports, they were not empty. Cross-sectionally (n = 272), others-directed meaning correlated with self-reported meaningfulness (WAMI: r = .24, p_{BH} = .002). So did agency (r = .20, p_{BH} = .011) and the contribution pathway (r = .23, p_{BH} = .005). In an adjusted model with all four dimensions, age, and occupation, the dimension block accounted for ΔR^2 = .08 in WAMI scores. The others-directed correlation held with narrative word count partialled (partial r = .21). It strengthened under graded two-coder scoring (r = .34). Random split-half re-estimation reproduced the block in all 200 resampled halves. The confirmatory nulls concern incremental prospective prediction, not basic convergence with self-report.

Location of the Concurrent Association in the Meaningfulness Distribution

Quantile regression (Koenker & Bassett, 1978) located the code–meaningfulness association in the lower tail of the meaning distribution (Figure 6). At the 25th percentile, slopes for agency and others-directed meaning (b ≈ 0.70) were about double their ordinary-least-squares counterparts. They shrank toward the mean effect at the 75th percentile. Narratives coded others-directed also showed lower meaningfulness variance (SD ratios .77–.79, Brown–Forsythe p_{BH} = .079). The association was asymmetric. Of respondents in the top WAMI quartile, 83% carried at least one codable meaning anchor. Only 30% of respondents with an anchor reached that quartile. A codable anchor was common among high-meaning respondents but was not sufficient

for high meaning. WAMI is ceiling-compressed (10.7% at the maximum). We interpret the shape of the association rather than the specific slopes.

Espoused but Not Narrated Meaning

Participants answered both an abstract question ("what makes your work meaningful") and a concrete critical-incident prompt. Codes assigned only from the abstract answer, and never grounded in the story, were unrelated to narrative code density ($r = .00$). They still predicted self-reported meaningfulness at both waves (Time 1 $r = .25$; Time 2 $r = .30$, both $p_{BH} < .01$). Espousing the significance of one's work without narrating it correlated positively with Time 2 turnover intentions ($b = 0.80$ [0.27, 1.33], $p = .003$, answer length partialled). Cell sizes are modest (48–65). The two prompts differ in format. Whether the pattern reflects claiming meaning one cannot ground is a question for a designed study.

Holding narrative richness constant, the number of facets on which the two coders disagreed was negatively related to concurrent meaningfulness ($b = -0.21$, $p = .004$, $n = 272$). Narratives from lower-meaning respondents were harder to code consensually. That is consistent with the low human–human agreement on communion facets reported in the Method.

A person-level profile analysis ordered mean meaningfulness across code configurations ($\eta^2 = .079$, $p_{BH} = .002$; Figure 7): agency + self + others > agency + others > agency + self > no codes. The gap opened where others-directed meaning joined agency. The analysis largely recapitulates the dimension-level results and is reported for completeness.

Screened Associations Not Retained

The breadth of the search bounds the retained results. No moderation of code–outcome associations survived screening (249 tests: 233 dimension × moderator interactions and 16 code × context interactions). The codes did not predict Time 2 retention. They did not predict job

satisfaction, affective well-being, or work–life interference or enrichment (0 of 35 tests). They were not reproducible from lexical style markers. An earlier 2017 coding of the same narratives carried no outcome signal. Huber re-estimation reproduced all five confirmatory coefficient estimates. Graded two-coder scoring did not add prospective incremental validity over Time 1 baselines (all $b^* \leq .06$).

No communion association survived the family-wise screen against Time 2 outcomes. The baseline-adjusted estimates in the confirmatory section were not robust to the unresolved-cell bounds. We note the directional consistency across four Time 2 outcomes as a pattern worth designed replication. Communion facets were also where human coders agreed least. We draw no conclusion from the pattern here.

Finally, the facet-level analyses were constrained by the codebook's resolution: six of thirteen facets occurred in five or fewer of the 276 narratives and could not be tested at all. The facet taxonomy exceeds what a sample of this size can support, which is itself relevant to the design of future narrative-coding studies.

Block 4's curvilinear predictions were withheld from formal testing before estimation. Only 4 of 276 respondents narrated three or more pathways (2 of them with Time 2 exhaustion data), so the density distribution cannot support a turning-point test. Figure S1 shows the descriptive quadratic estimates, scored as signs only; the corresponding concurrent quadratic on Time 1 meaningfulness appears in Figure S2. The overall quadratic for density on Time 2 exhaustion was positive as predicted ($b_2 = 0.21$, 95% CI [−0.18, 0.60], $p = .288$). The quadratics on both Time 2 meaning outcomes were negative as predicted. The comparative prediction (Hypothesis 4b) was uninformative. Both orientation-specific quadratics were negative and imprecise (other-oriented $b_2 = -0.30$, $p = .209$; self-oriented $b_2 = -0.11$, $p = .813$). Almost no

respondents sat at the density levels where curvature would appear. The quadratic intervals include zero.

Study 2: Meaning Events in Occupational Discourse

Corpus

The corpus is Internet Archive snapshots of allnurses.com captured between 1 January 2016 and 8 September 2026 (2026 is therefore a partial year, 147 posts): 7,695 posts of at least 200 characters, in 1,453 threads, by 4,442 hashed member accounts (1,670 threads had cached HTML; recoverable boards are listed in Appendix D, Table D1). The scoring file held 7,709 successful records; 14 posts had been captured and scored twice, and the first record for each post was kept, leaving 7,695 unique posts. Capture year is not posted date; we use three capture windows (2016–2019, 2020–2022, 2023–2026) as a proxy for period. Posts are nested in threads and in accounts; the 34% of accounts that posted twice or more wrote 62% of posts. All intervals are 95% percentile intervals from a bootstrap that resamples threads (B = 2,000); account-level resampling gave intervals of the same or narrower width for every period contrast reported.

Coding Instrument

Each post was coded by a single large language model (claude-sonnet-5, one pass at the API default sampling temperature; the temperature lock specified in the protocol was not applied in the scoring client, a deviation we record here; prompt frozen and hashed before the run) into zero or more meaning events. An event is a passage in which a Rosso et al. (2010) facet of meaning is enacted, frustrated, or wished for. Each event carries the facet, its polarity, the object implicated (a named AI system; a system implementation or rollout; an optimization practice such as staffing ratios, productivity metrics, dashboards, or documentation load; or none), the writer's stance toward that object, what became of the task involved (taken, added, reconfigured,

unchanged), the emotion carried, and to whom the writer attributes the situation (the tool, management, staffing, the patient, the self). Four post-level flags record expressed intent to leave, depressed affect, anxious affect, and a sense of lacking control.

No human coding of forum posts was undertaken, and no second-model rescoring, prompt-perturbation check, or negative-control set has been run for this instrument. Its agreement with human judgment is known only from Study 1's participant narratives, a different genre in which the same prompt family did not meet the acceptance standard. The false-present rate on forum prose is therefore unknown, and every rate below is a rate of model-assigned codes. We treat this as the principal measurement limitation of Studies 2 and 3.

A closed-form dictionary layer (care, mastery, optimization, AI/automation, leaving or strain) was retained as archive description only. Its stems are occupational nouns: on a nursing board, a hit on the care list mostly records that a post is about nursing. We report it where it bears on coverage (how often AI vocabulary appears at all) and do not read its year rates as facet scores.

Career-Point Indicator

The allnurses author pane displays the credential line members choose to publish (e.g., "BSN, RN," "MSN, APRN, NP," "LPN," "Nursing Student"). We parsed that pane, never the post body, into a seven-level role class (student; aide/technician; LPN/LVN; RN; advanced practice; educator/manager; unclassified). A classifiable badge was present for 48% of posts; coverage varied with the page template the archive happened to serve (69% under the 2016–2018 template, 39% under the later one), so role is analyzed as a stratifier within period rather than as a population share, and role-by-year tables are confounded with markup. The badge reflects the credential at the time the page was captured, not necessarily at the time of posting.

Results

The coder returned 5,429 events (0.71 per post); 66% of posts carried no codable meaning event, which is expected in a forum where many posts are short replies. Events per post fell in 2019 (0.44) and 2021 (0.52), with 2020 at 0.66, versus 0.70–0.93 in the other capture years, the same years in which the site's page template changed and mean post length dropped; that concurrent change conditions every trend statement in Study 3.

Nurses rarely referred to AI. Across ten years and 7,695 posts, the coder found seven events in which a named AI system was the object, all in pages captured in 2020–2021 and none in 2023–2026; AI vocabulary from the dictionary layer appeared in under 1% of posts in any year. The objects nurses did name were the instruments by which their work is measured and reorganized: of 629 object events, 73% concerned an optimization practice (ratios, productivity targets, dashboards, documentation load) and 26% a system implementation or rollout. About one post in twenty carried such an object in every period (Figure S3).

Figure S4 shows events per 100 posts by facet, polarity, and period. Two facets carried most of the agency signal in every period: Competence, the most frequently enacted facet (10.7–11.6 enacted events per 100 posts outside the 2020–2022 trough), and Control/Autonomy, the most frequently frustrated facet. The communion-family facets (Dyad Connectedness, Social Identification, Value Systems, Interconnection, Self-Abnegation) together accounted for 22% of all events, more than Competence, but were distributed across five rows none of which exceeded 4.4 per 100 posts, and they were predominantly enacted (650 enacted, 444 frustrated, 92 wished for). Half of all object events (49%) targeted Control/Autonomy and a further 21% Competence. Among the 629 object events, about three-quarters were frustrated overall (74%; 66% in 2020–2022) and about three-quarters carried a dissatisfied stance (76%, 71%, and 79% by period)

(Figure S5). Attribution went to management in 51% of events and to staffing in 22%; the tool itself was blamed in 13%. These are shifts of roughly ten to thirty events per period cell and are reported as description, not as tested contrasts.

Posts that discussed an optimization or implementation object carried an expressed intent to leave at 13.8%, 14.9%, and 15.9% in the three periods, against 1.6%, 1.7%, and 1.9% among other posts (differences 12.1 pp [5.5, 19.3], 13.2 pp [3.9, 22.2], 13.9 pp [6.2, 22.6]); the lack-of-control flag was set in 64–73% of object posts and 7–11% of others (Figure S6). The reverse conditional probabilities qualify this result: pooled across periods, 30% of quit-flagged posts named an object [95% CI 21, 39], so seven in ten did not; 28% of lack-of-control posts did [24, 33]; and 85% of object posts carried no quit flag. Two features of this contrast limit what it can mean. The codebook permits the lack-of-control flag to co-occur with a frustrated Control/Autonomy event, so the first row is in part one complaint counted twice; and posts that raise an optimization object are, by construction, complaint posts, so the comparison group includes questions, advice, and reminiscence. Restricted to the 1,317 posts with at least one frustrated event, the quit-intent difference between optimization complaints and other complaints shrank to 7.5 pp [1.9, 12.9] pooled (17.6% vs 10.1%) and did not exclude zero within any single period. The association is a same-post co-occurrence coded by one model in every period; it is not a temporal ordering and is not evidence that optimization talk precedes exit.

Dictionary and model layers agreed weakly at the post level: optimization dictionary density with any-object code, $r_{pb} = .17$; leaving-strain density with the quit-intent flag, .16; AI dictionary hits with named-AI events, $\varphi = .13$ (4% of dictionary hits carried a named-AI event). Event count correlated with post length ($r = .33$). Communion vocabulary was three times denser in posts carrying any event (0.36–0.41 hits per 100 words) than in posts carrying none (0.13),

and about equally dense whether the event was a Control frustration or another frustration; 36% of Control-frustrated posts also carried a communion-family event.

A coding scheme developed on structured survey narratives does not automatically transfer to unstructured, self-selected posts, and no transfer sample exists. We treat Study 2 as a completed field description of model-assigned codes with those limits stated.

Study 3: Change in Meaning Talk Across Capture Years

Study 3 tested whether the talk described in Study 2 moved once generative AI entered the occupational conversation. The time axis remains Wayback capture year, and the 2019–2021 template change conditions every series (Figure S3). Period contrasts (2023–2026 versus 2016–2019, and versus 2020–2022) use thread-cluster percentile bootstrap 95% confidence intervals (B = 2,000) and two-sided thread-level permutation tests of period labels (B = 4,000), with Benjamini–Hochberg screening at q = .10 within each family (post-level shares; facet-by-polarity cells; object-event category shares; badge strata); all are descriptive screens, not pre-specified tests.

The share of posts carrying any optimization or implementation object was flat: 4.8% in 2016–2019 and 5.2% in 2023–2026 (difference 0.4 percentage points, 95% CI [−1.8, 2.5]; linear slope 0.00 pp per year [−0.27, 0.27]; 0.01 [−0.26, 0.28] adjusting for log post length). Named-AI events were zero in the 2023–2026 window. Neither the share of posts attending to optimization nor the presence of AI as a named object changed over the decade.

The change was in expressed control. The lack-of-control flag rose from 10.5% of posts to 14.5% (difference 4.0 pp [1.2, 6.7]; slope 0.49 pp per year [0.13, 0.89]; 0.50 [0.14, 0.85] adjusting for log post length). Frustrated Control/Autonomy was the only facet cell that grew across periods (9.6, 9.8, 12.7 frustrated events per 100 posts; 2023–2026 versus 2016–2019

difference 3.1 [0.5, 6.0], permutation $p = .027$), but the difference did not survive Benjamini–Hochberg screening across the 20 facet-by-polarity cells ($q = .27$; Table S3). Posts with at least one frustrated event rose from 16.2% to 19.8% (3.5 pp [0.2, 6.6]), but the year series shows this as recovery from the 2019–2021 trough to a level near 2016's 18.6%, and the linear slope did not exclude zero (0.35 pp per year [−0.11, 0.79]). We therefore describe rising control loss with the template caveat attached, and we do not describe rising frustration. Quit-intent, anxious, and depressed flags showed no trend. Across the eight post-level shares, the lack-of-control flag was the only contrast that survived screening ($q = .028$). Against 2020–2022, seven enacted facet cells rose ($q < .10$), which is the recovery from the template-change trough, not a change dated to 2023.

Within object events, self-attribution fell from 16% in 2016–2019 to 6% in 2023–2026, and the share of events in which the task was described as reconfigured (15% to 22%) or added to (5% to 9%) grew while "taken" stayed at 4–5%. These are cell shifts of tens of events: the self-attribution decline was −9.7 percentage points [−19.5, −1.2] (permutation $p = .046$, $q = .46$ across 19 category shares), and the task-outcome shifts did not exclude zero. RN-badged posts showed no change in object exposure (5.5% to 5.4%; −0.1 pp [−3.0, 2.7]); advanced-practice posts rose from 3.3% to 6.7% (3.4 pp [−0.2, 7.2], permutation $p = .11$); LPN/LVN cells were small, and student and aide posts were too few after 2019 to estimate (Figure S7). Because the badge is read at capture rather than at posting, the advanced-practice change is compatible with career progression among long-standing members as well as with rising exposure of that role.

Exposure remains the principal limitation. Capture year is not use of an AI system, and hospital- or community-level adoption would not establish that any particular writer was affected. The cross-period comparison is also among people who chose to post on one professional forum;

what changed may be who posts as much as what nurses experience. We report the year contrast as a completed description of the available window.

Discussion

The studies asked which coded kinds of meaning, if any, sustain helping work and whether that map is visible in occupational talk after generative AI arrived. The pre-specified tests placed the year-ahead associations between narrated meaning dimensions and felt meaningfulness, exhaustion, and turnover intentions near zero; for next-year meaningfulness the Bayes factor (BF01 = 481.5) favors the null decisively. The same codes were associated with self-reported meaningfulness in the same wave, which supports the concurrent validity of the coding scheme. One prospective association survived screening: when Time 1 stories grounded meaning in competence or mastery, Time 2 turnover intentions and emotional exhaustion were higher. That last result is exploratory and rests on 17 present-coded cases. It was also the only coded feature that out-predicted the Time 1 survey battery for strain. Two interpretations are consistent with the result. Meaning that must be earned through demonstrated skill can become a contingent investment. Meaning staked on mastery can also convert to restlessness when the job stops supplying growth. Those readings treat the code as an anchor in the story. Whether the underlying need was met is a different question; Martela and Riekki (2018) locate meaningfulness in competence need satisfaction, not in narrative mention of mastery alone.

Codes assigned from the abstract meaning answer but absent from the story were orthogonal to the story codes. They still predicted self-reported meaning. The meaning literature already distinguishes claiming or seeking meaning from its presence; the former often tracks

distress rather than adjustment (Steger et al., 2006; Park, 2010). Espousing the significance of one's work while narrating none of it may reflect that dissociation. Prompt format remains an alternative explanation.

The dark-side conjecture that motivated Block 4 (Rosso et al., 2010) could not be tested. Almost no one narrated more than two pathways. The descriptive quadratics on exhaustion and on the two Time 2 meaning outcomes carried the predicted signs. The comparison of other-oriented and self-oriented curvature was uninformative. Whether the cost of densely invested meaning is communal or agentic remains open.

Objects of Frustrated Meaning in Nurses' Discourse

Studies 2 and 3 asked how a decade of nurses' own talk about their work relates to the account of the AI paradox developed in Study 1. The first result is an informative absence. Nurses rarely referred to artificial intelligence: seven events in 7,695 posts named an AI system, none in the 2023–2026 window, and AI vocabulary appeared in fewer than one post in a hundred in any year, including the years after generative tools entered wide public use. A study that operationalized AI exposure as mention of AI would have detected nothing and would have misattributed the reason.

What nurses discussed, in a steady one post in twenty across ten years, were the instruments that measure and reorganize their tasks: ratios, productivity targets, dashboards, documentation load, and the arrival of new systems. We interpret this as the phenomenon the AI-paradox literature describes, in the terms practitioners use for it. The paradox as formulated here is not about a technology but about a mode of organizing work in which decisions about how a task is done migrate from the person doing it to a system that scores it. Nurses have described that mode since before the current wave of tools, and their vocabulary for it is managerial and task-focused,

not technological. The implication for measurement is direct: exposure to algorithmic management in occupational text must be sought through the artefacts and practices themselves, not through the word "AI," which is a researcher's category rather than a worker's. We do not claim from these data that AI is eroding meaning, or that any Rosso facet declined after 2023; the data show neither.

The second result concerns which facet carries the strain. Across periods the cell that grew was frustrated Control/Autonomy, the flag that rose was lack of control, and half of all object events landed on Control/Autonomy. Competence remained the most enacted facet in every period, and the communion-family facets were enacted far more often than frustrated. In Rosso et al.'s (2010) terms, where an optimization object is named, the facet at issue is discretion over means, not skill. Study 1's pre-specified agency paths were estimated near zero, so the prospective evidence available here is the one exploratory association that survived screening, and it runs from competence toward strain, not protection. What Studies 2 and 3 add is descriptive.

Communion deserves a separate comment because it is both pervasive and hard to code, and those facts are related. Communion vocabulary saturates the posts that carry any meaning event, but it rarely appears as a discrete episode with a polarity. A sentence in which a nurse cannot give patients the care they deserve has care as its object and the blocked action as its marked feature; both human coders in Study 1 and the model here register it as frustrated Control, while the communion it presupposes goes uncoded. We suggest this is why inter-coder agreement on Communion sits near .5 for humans and models alike (Study 1) and why the facet is dispersed across five low-frequency facets here. That reading yields a hypothesis rather than a finding: the complaint arrives framed as autonomy loss because the injury is to communion, and autonomy

over one's time is the means to it. Testing it requires a measurement approach that treats communion as a property of the post rather than of the event.

Practical Implications

Because Studies 2 and 3 are same-post co-occurrence designs in a self-selected venue, their implications concern measurement rather than intervention. The relevant exposure in nursing is not the presence of an AI tool but the manner in which any system arrives: as a metric, a new documentation duty, or a management mandate. Screening a deployment for what it does to the task (taken, added, reconfigured) and to discretion over means is consistent with the pattern here in a way that screening for "AI" is not. The sense of lacking control is the only strain marker that moved over the decade; its co-occurrence with optimization talk in the same post is partly a property of the codebook and is not evidence of temporal order, so whether discretion over means is an earlier signal than burnout inventories is a question for a lagged, within-account design that has not been run. Competence and relational connection are the meaning resources these data show still being enacted; deployments that displace skill or compress patient contact would act on the parts of the job that remain intact.

Boundary Conditions

Several boundary conditions delimit the paradox. Meaning is heterogeneous. A task that is meaning-critical for one worker may be routine for another (Tang et al., 2022). The paradox is a claim about distributions of meaning across a workforce, not a universal task ranking. Economic context also matters. In precarious conditions the most consequential feature of work may simply be having a job. The diseconomy of scale we describe becomes decision-relevant when efficiency and retention are both at stake. Finally, the claim is about meaning-critical tasks.

Where automation removes genuinely low-meaning work, the efficiency case is intact (Albrecht et al., 2025; Olson et al., 2025).

Strengths and Limitations

Study 1 establishes a meaning-criticality map, not a measured AI effect. Its strengths are a year-lagged design, confirmatory tests at the more reliable dimensional level, and an attempt to test previously untested conjectures. Its limitations are equally clear. The Time 1 data predate the automation wave, so the paradox is read by juxtaposition. The sample is an online panel and is majority, but not exclusively, health care. Studies 2 and 3 widen the text base. They do not resolve individual-level exposure. We prefer that claim to a causal one the design cannot support. Studies 2 and 3 carry their own limits. The instrument is a single model applied once, without human coding of forum text and without second-model, perturbation, or negative-control checks, so the false-present rate on forum prose is unknown. The frustration series is confounded with a page-template change that shortened posts and reduced codable events in 2019–2021; we report rising control loss with that caveat and do not claim rising frustration. Period is capture period, not posting period. The exposure–strain contrast compares complaint posts with all other posts and is partly circular by codebook design; most quit-flagged posts name no object. The career-point badge is a capture-time credential with template-dependent coverage; aides and students are sparse in the venue after 2019. The dictionaries are occupational vocabularies, not facet measures, and we draw no year-trend conclusion from them. A within-account design over the 1,494 accounts with two or more posts, using posted rather than captured dates, is the analysis that would move "indicator" from a suggestion to a claim; it has not been run.

The engineered prompts fell short of an acceptance standard tied to human–human agreement, a result about this codebook and prompt set rather than a ceiling on machine coding.

Human agreement on communion and self-directed facets was itself low. Prompts selected to match those labels can transmit that disagreement into the scores. An independent panel that applies the same scheme without being fitted to a particular human pair, then evaluated against later well-being and retention rather than against human PSA, may give more accurate estimates than either the disqualified prompts or the human composite. More accurate estimates may be more predictive. That test is the next measurement study, not a result of this article.

A further limit is that the coding scheme records presence, not salience. Human coders made binary judgments. No graded measure of how central an anchor was to the story exists in these data. The one machine attempt at such a measure did not meet the pre-set reliability bar. Dichotomization at low base rates attenuates associations. Under a normal liability model, the observed point-biserial correlations of .28 between competence codes and Time 2 strain imply latent continuous associations of .44 to .46 (Figure S8). Those biserial estimates assume latent normality and are illustrative. They suggest that the binary results are more likely to understate than overstate the underlying continuous associations.

The pre-specified tests, the concurrent convergence, and the competence–strain association constitute the findings. The remaining screened tests bound how far those three can be interpreted.

Conclusion

In these data, no global meaning score forecast turnover intentions; the one prospective association involved the kind of meaning narrated, specifically competence, and next-year strain. The optimization case for automating helping work still assumes a known split between routine and meaningful tasks. The pre-specified dimensional map did not forecast next-year outcomes; its estimates were bounded near zero. A decade of nurses' own talk named the instruments of

optimization, not AI, and what rose in it, with the template-change caveat attached, was a sense of lacking control over means. The competence result is the one that carried prospective signal, and it is the result a designed study should attempt to falsify.

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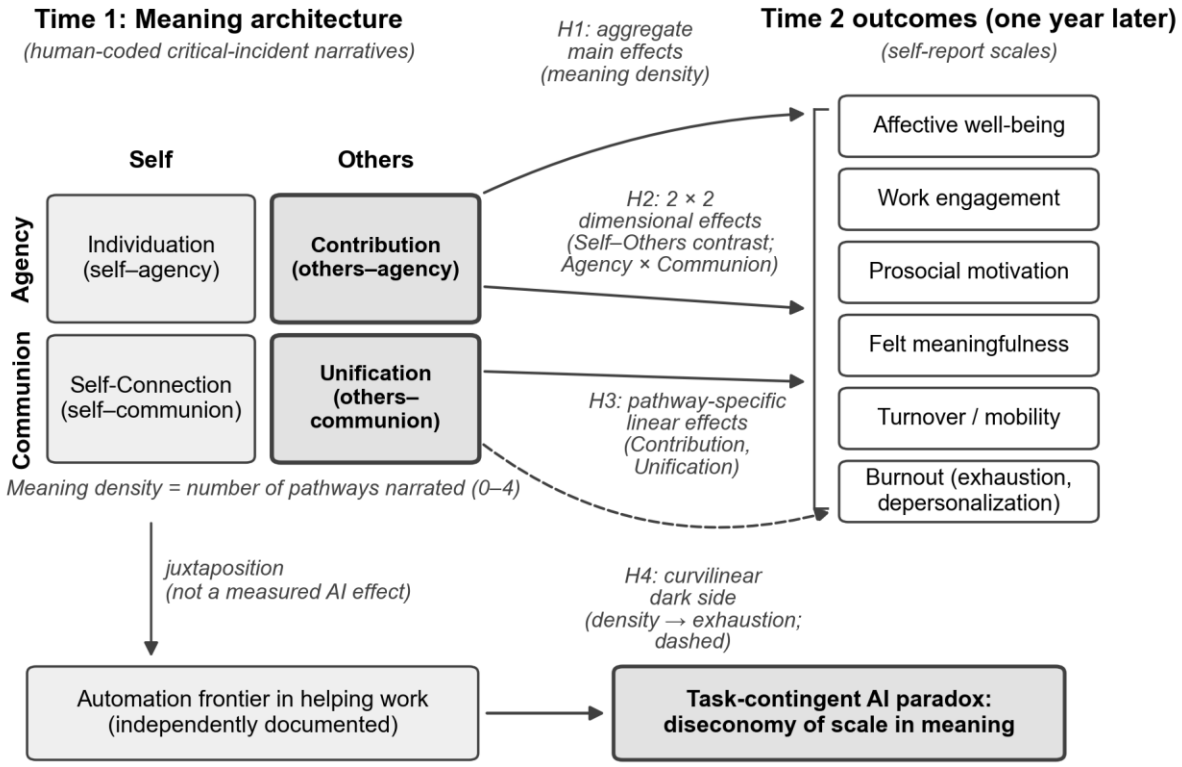
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Figure 1

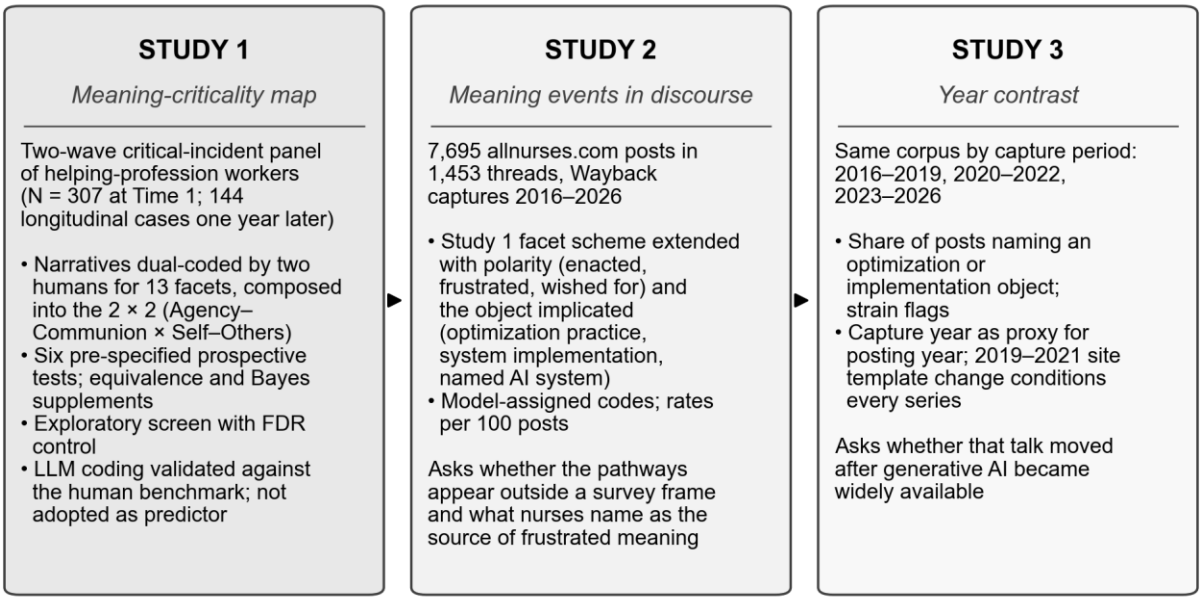
Conceptual Model: Time 1 Meaning Architecture, Time 2 Outcomes, and the Paradox



Note. The 2 × 2 organizes meaning by Agency–Communion (rows) and Self–Others (columns); meaning density is the number of pathways a narrative enacts (0–4). Arrows denote the hypothesis blocks: aggregate main effects of density (H1), 2 × 2 dimensional effects (H2), pathway-specific linear effects of Contribution and Unification (H3), and the curvilinear dark-side effect of density on exhaustion (H4; dashed). Other-oriented pathways (bold) carry the paradox-relevant meaning. The task-contingent artificial intelligence (AI) paradox is inferred by juxtaposition with the independently documented automation frontier, not from a measured AI effect. This article reports three of the six Time 2 outcomes: felt meaningfulness, emotional exhaustion, and turnover intentions (Table 2).

Figure 2

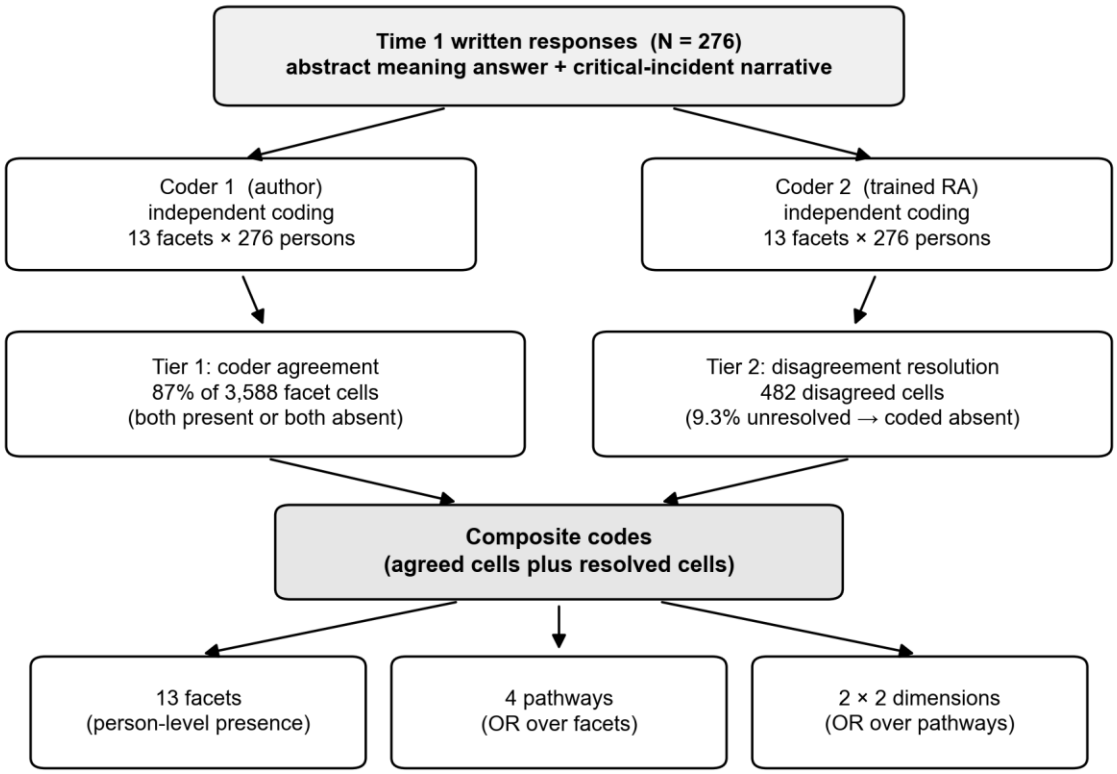
The Three-Study Research Program



Note. Study 1 establishes the meaning-criticality map from human-coded critical-incident narratives. Study 2 applies the same facet scheme, extended with polarity and the object implicated, to a decade of nursing-forum posts; Study 3 contrasts that talk across capture periods. The AI paradox is read by juxtaposing the Study 1 map against the independently documented automation frontier. No study measured individual AI exposure. *N* = sample size; FDR = false discovery rate; LLM = large language model; AI = artificial intelligence.

Figure 3

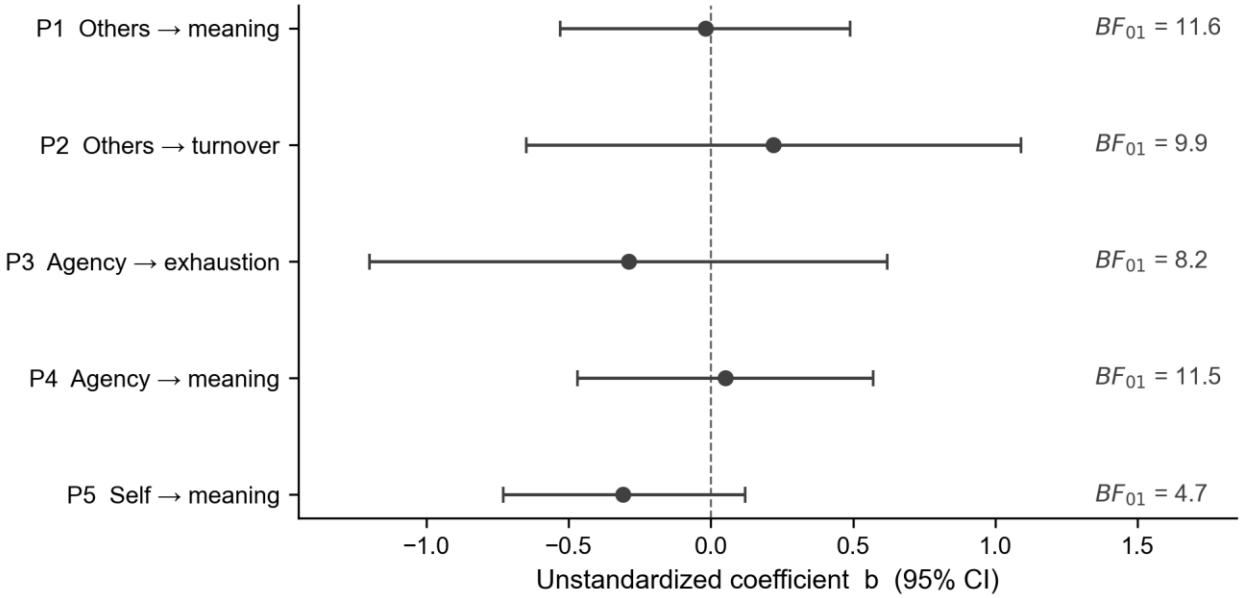
Two-Tier Human Coding and Disagreement Resolution



Note. N = 276 persons. PSA = proportion of specific agreement (Cicchetti & Feinstein, 1990), equivalent to F1 (van Rijsbergen, 1979). Unresolved Tier-2 cells (9.3%) were coded absent; an endpoint sensitivity check was run. Dimension prevalence: Agency 64%, Others 56%, Self 23%, Communion 11%. Dimension-level human–human PSA: Agency .935, Others .844, Self .52, Communion .48. RA = research assistant; OR = present if any constituent facet (or pathway) is present.

Figure 4

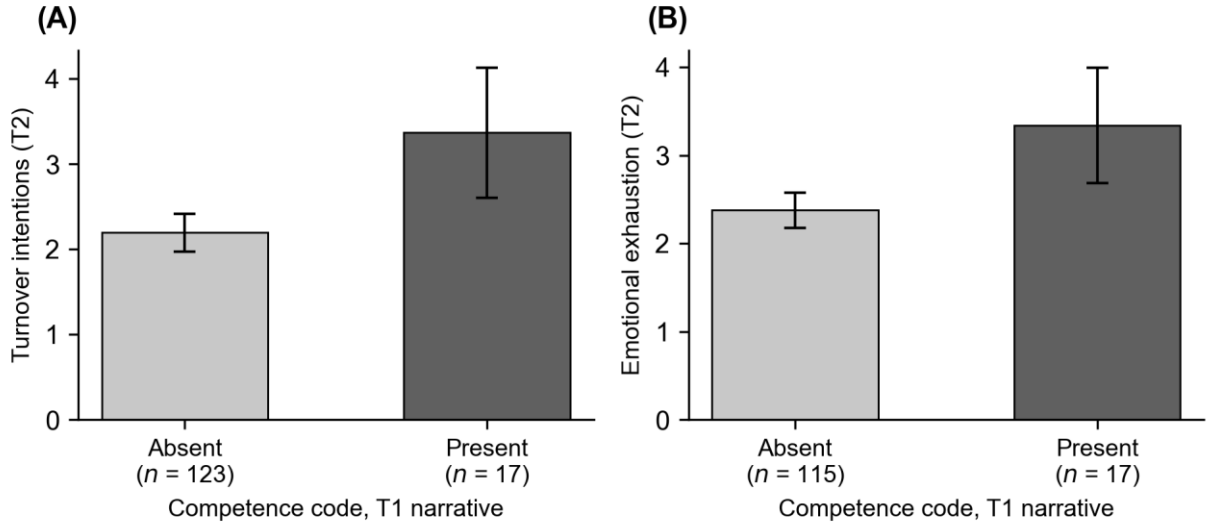
Pre-Specified Prospective Estimates



Note. Error bars are 95% confidence intervals for the unstandardized coefficients reported in Table 2. P6, the joint block test, has no single coefficient and is reported in Table 2 only. n = 136 (P3 n = 128). BF_{01} is the BIC-approximated Bayes factor (Wagenmakers, 2007) favoring the null (Table 3).

Figure 5

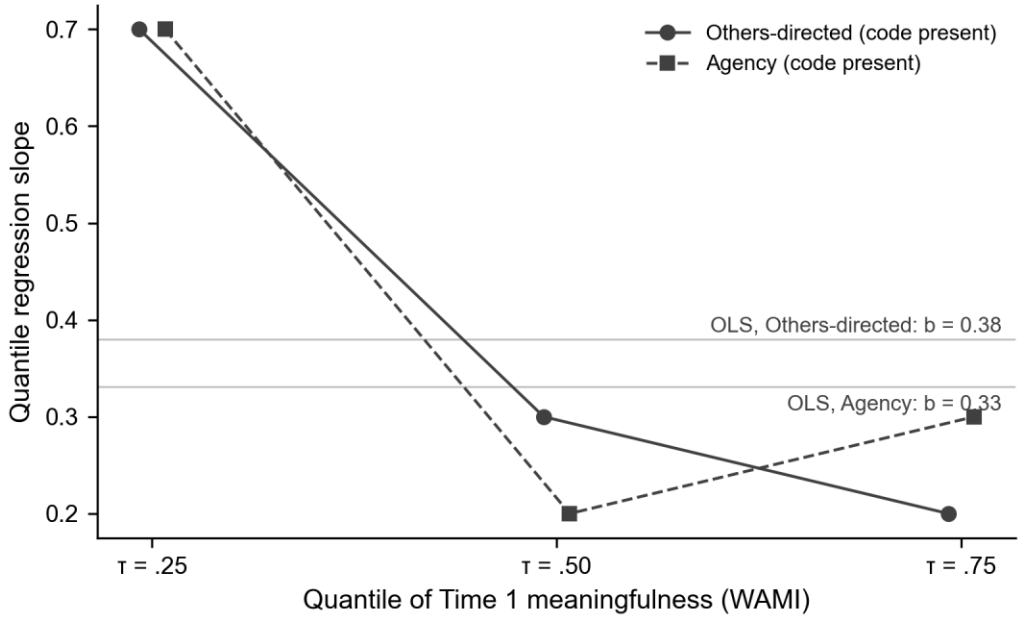
Time 2 Strain by Time 1 Competence Code



Note. Unadjusted cell means with 95% confidence intervals. Panel A: turnover intentions (n = 123 absent, 17 present). Panel B: emotional exhaustion (n = 115 absent, 17 present). Adjusted estimates are reported in the text and in Table 4. Exploratory.

Figure 6

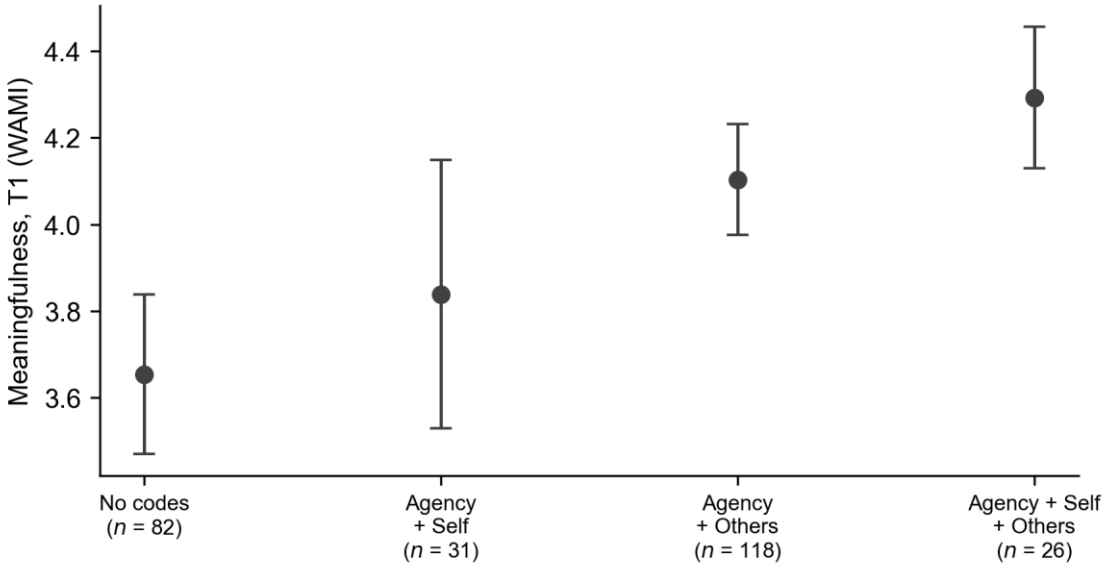
Quantile Regression Slopes for Time 1 Meaningfulness



Note. Markers show slopes of Time 1 meaningfulness (Work and Meaning Inventory; WAMI) on each binary code at the .25, .50, and .75 quantiles (n = 272). Horizontal lines are the corresponding ordinary least squares slopes, labeled at the right. τ = quantile; OLS = ordinary least squares; b = unstandardized slope.

Figure 7

Time 1 Meaningfulness by Narrative Code Configuration



Note. Error bars are 95% confidence intervals. WAMI = Work and Meaning Inventory. Configurations are discrete; cells match the exploratory configuration analysis. The four plotted cells total n = 257 of 276 coded narratives; 19 narratives in rarer configurations are omitted.

Appendix A

Meaning Coding Framework

Table A1

Hierarchy of the Coding Scheme: 13 Facets Nested in Four Pathways and a 2 × 2 Superordinate Structure

Pathway	Agency–Communion	Self–Others	Constituent binary facets
Individuation	Agency	Self	Control/autonomy; competence; self-esteem
Contribution	Agency	Others	Perceived impact; significance; interconnection; self-abnegation
Self-Connection	Communion	Self	Self-concordance; identity affirmation; personal engagement
Unification	Communion	Others	Shared value systems; social identification; dyadic connectedness

Note. Facets are coded present/absent on each narrative and aggregated upward by an inclusive (logical-OR) rule. The 2 × 2 superordinate dimensions are Agency = Individuation + Contribution; Communion = Self-Connection + Unification; Self = Individuation + Self-Connection; Others = Contribution + Unification. Labels follow the project coding instructions adapted from Rosso et al. (2010). The interconnection and self-abnegation facets reflect transcendence; the seven narrative mechanisms in Rosso et al. do not form a coding layer here.

Appendix B

Summary of Hypotheses

Table B1

Time 1 → Time 2 Hypotheses by Block, Predictor, Outcome, and Predicted Form

H	Block	T1 predictor	T2 outcome	Predicted form
1a	Aggregate	Meaning density (breadth)	Well-being, engagement	Positive linear
1b	Aggregate	Meaning density	Prosocial motivation	Positive linear
1c	Aggregate	Meaning density	Turnover/mobility	Negative linear
1d	Aggregate	Meaning density	Felt meaningfulness	Positive linear
2a	Dimensional 2×2	Self–Others contrast	Well-being, engagement	Others > Self
2b	Dimensional 2×2	Agency × Communion	Outcomes (composite)	Non-additive interaction
2c	Dimensional 2×2	Self–Others contrast	Depersonalization vs. exhaustion	Differential by subscale
3a	Pathway-specific	Contribution	Well-being, engagement, prosocial	Positive linear
3b	Pathway-specific	Unification	Engagement, prosocial; turnover	Positive; negative (turnover)

H	Block	T1 predictor	T2 outcome	Predicted form
3c	Pathway-specific	Individuation, Self-Connection	Sustaining outcomes	Weaker/less consistent
4a	Curvilinear	Meaning density	Burnout (exhaustion)	U-shaped past turning point
4b	Curvilinear	Other- vs. self-oriented density	Burnout	Stronger curvature (other-oriented)
4c	Curvilinear	Meaning density	Affective well-being	Inverted-U

Note. Predictors are coded from Time 1 (T1) critical-incident narratives; outcomes are measured at Time 2 (T2), approximately one year later. Blocks 1–3 specify linear, workforce-sustaining relationships; Block 4 tests the curvilinear conjecture (Rosso et al., 2010, p. 116) following Haans et al. (2016). Hypothesis 1d links the coded meaning predictor to the felt-meaningfulness outcome (Monnot & Beehr, 2014) as distinct constructs. Confirmatory tests are specified at the 2 × 2 level; pathway- and facet-level analyses are secondary.

Appendix C

Measures Administered by Wave

Table C1

Constructs, Wave of Administration, and Source Instruments

Construct	Wave	Source / instrument
Job-meaning description (open-ended)	T1	Study-specific prompt
Critical-incident narratives (context, event, outcome)	T1	Critical incident technique (Flanagan, 1954)
Work engagement	T2	Breevaart et al. (2012)
Prosocial motivation	T2	Grant (2008a)
Affective well-being	T2	Warr et al. (2014)
Meaningfulness of work	T1 and T2	Monnot & Beehr (2014); reverse items: irrelevant, trivial
Burnout (higher-order; exhaustion, depersonalization)	T2	Maslach et al. (2001)
Turnover intentions / voluntary mobility	T2	Study-specific items
Meaningful work (WAMI, 10 items)	T1 and T2	Steger et al. (2012); 1 = absolutely untrue to 5 = absolutely true

Construct	Wave	Source / instrument
Work centrality (1 item)	T1	Single item; covariate

Note. Meaningfulness is a self-report outcome distinct from the coded predictor.

Appendix D

Boards Represented in the Allnurses Archive

Study 2 and Study 3 draw on Internet Archive snapshots of allnurses.com, not a complete crawl of the live site. The enumerator covered the domain for capture years 2016–2026. The analyzed set is a stratified thread sample (1,566 threads with cached HTML at the time of the board census; the final corpus has 1,670). Board names were recovered from page breadcrumbs or JSON-LD BreadcrumbList strings that appeared in at least five threads. Later allnurses URLs no longer carry the forum in the path, so 661 threads have no recoverable board label. A thread can carry more than one label when a parent section and a child board both appear. The intended primary frame was working-nurse discussion. Student and school boards remain in the sample because the URL filter could not see them on later pages. Table D1 reports the recoverable boards.

Table D1

Boards Recovered in the Fetched Allnurses Sample

Group	Board	Threads
Advanced practice	Nurse Practitioners	16
Career	Nursing Career Advice	73
Geography	United States Nursing	91
Geography	International Nursing	30
Geography	New York Nursing	13
Geography	California Nursing	12

Group	Board	Threads
Geography	Florida Nursing	7
Geography	Arizona Nursing	6
Geography	Texas Nursing	5
Specialty	Ob/Gyn Nursing	19
Specialty	Emergency Nursing	17
Student / school	Nursing Students	261
Student / school	General Student Support	56
Student / school	Pre-Nursing Students	44
Student / school	NCLEX Exam and Programs	32
Student / school	Nursing Student Assistance	17
Student / school	Student Nurse Practitioner	12
Support role	CNA / Medical Assistant	12
Working-nurse discussion	Nursing Specialties	224
Working-nurse discussion	General Nursing	194
Working-nurse discussion	Nurse Support	162
Working-nurse discussion	New Nurses / First Year After Licensure	23

Group	Board	Threads
Unlabeled	Forum not recoverable from the snapshot	661

Note. Cell entries are thread counts in the fetched sample, not post counts and not the full Internet Archive frame. Labels are public board names. Hapax breadcrumb strings (almost always thread titles) are omitted. Student/school boards were specified as outside the primary frame and are shown so the mix is visible. 2026 coverage in the current frame is thin.

Supplemental Material

Table S1

Prompt-Level Results of the Pre-Specified LLM Coding Validation

Prompt	Type	Rank	Agency	Others	Communion	Self	FPs	Error	ICC
Decision-first	single call	.808	.798	.819	.531	.389	28	.018	.58
Per-facet	13 calls	.790	.780	.800	.583	.382	24	.000	.23[a]
Reason-first	reason then decision	.787	.781	.793	.545	.333	25	.036	.62
Decision-first, revised	single call, revised	.781	.755	.806	.558	.353	22	.000	.60
Decision plus evidence	decision plus evidence fields	.764	.751	.777	.549	.378	30	.000	.46[a]
Alternate codebook	alternate codebook	.756	.750	.761	.471	.262	13	.036	.49[a]

Note. Rank = mean of agency and others-directed PSA (fixed before the evaluation). Acceptance thresholds: agency ≥ .910; others-directed ≥ .805 (lower bounds of the human–human benchmark bootstrap 95% CIs). Agency, Others, Communion, and Self columns are positive specific agreement (PSA) with the composite gold. FPs = pooled false positives across three replicates on communion facets and on sparse facets over agreed-absent cells; any such false positive disqualified. Error = final-attempt structured-output failures / 165 case-calls. ICC = salience ICC(2,1); verdicts follow the pre-declared thresholds (reportable ≥ .75; exploratory ≥

.50); salience was non-voting. None of the six prompt configurations met the agency threshold (.750–.798 against ≥ .910). The human composite codes were retained. [a] Below the exploratory band; salience was dropped for that prompt.

Table S2

Descriptive Test-Subsplit Estimates for the Two Highest-Ranked Disqualified Prompts

Comparison	Prompt	Agency	Communion	Self	Others
Coder 1 vs. coder 2 (13 facets)	—	.95 [.91, .98]	.53 [.20, .77]	.50 [.32, .66]	.84 [.76, .91]
Model vs. composite gold	Decision-first	.81 [.71, .89]	.45 [.12, .71]	.53 [.33, .70]	.746 [.626, .847]
Model vs. composite, excl. rule-generating cases	Decision-first	.80 [.70, .88]	.48 [.12, .73]	.50 [.29, .67]	.76 [.64, .86]
Model vs. coder 1 (13 facets)	Decision-first	.84 [.77, .90]	.27 [.04, .49]	.57 [.41, .70]	.75 [.65, .83]
Model vs. coder 2 (13 facets)	Decision-first	.86 [.79, .91]	.39 [.10, .63]	.42 [.23, .59]	.81 [.73, .88]
Model vs. composite gold	Per-facet	.80 [.70, .88]	.39 [.00, .67]	.49 [.26, .68]	.77 [.66, .86]
Model vs. composite, excl. rule-generating cases	Per-facet	.81 [.71, .89]	.43 [.00, .73]	.47 [.23, .67]	.79 [.67, .88]
Model vs. coder 1 (13 facets)	Per-facet	.85 [.78, .91]	.31 [.00, .56]	.60 [.43, .74]	.75 [.65, .83]
Model vs. coder 2 (13 facets)	Per-facet	.87 [.80, .92]	.33 [.00, .60]	.41 [.21, .59]	.84 [.77, .91]

Note. Test-A is the pre-specified 65-case subsplit of the 193 sealed test cases; the protocol manifest was hashed before scoring. Cells are positive specific agreement (PSA) with case-cluster bootstrap 95% confidence intervals (B = 10,000). Primary model claude-sonnet-5, three replicates pooled; structured-output failures left 64, 64, and 63 of the 65 cases scored across the three decision-first replicates (pooled n = 64, n = 61 excluding rule-generating cases); the per-facet prompt scored all 65 in every replicate. gpt-5 ran non-voting and completed 65 cases under the decision-first prompt but 22 of 65 under the per-facet prompt, so its per-facet pass is incomplete and is not tabled. Coder-vs-coder and model-vs-coder rows use all 13 facets, the quantity the published human benchmark reports (.93/.84 on 276 cases); composite rows use the gold-labeled facet set, which is what the scoring pipeline evaluates; the two are not equivalent.

Pre-specified selection verdict on test-A: both prompts disqualified on false positives over agreed-absent cells (decision-first: 33 across seven facets; per-facet: 33 across five); structured-output error rate 2.1% (decision-first) and 0% (per-facet). Pre-specified adequacy screen for scoring the remaining 128 cases (decision-first pooled Agency and Others both ≥ .75): Agency = .809, Others = .746; the screen was not met, so the 128 cases remained sealed under the pre-specified protocol; a single, separately authorized pass over those cases is not reported here. The Others cell for that row is shown to three decimals because it sits at the threshold. Salience ICC(2,1): decision-first .43, per-facet .09, both below the exploratory band. Rule-generating cases = the 4 test-A cases whose blinded adjudications generated codebook decision rules; excluding them moved no estimate by more than .04. Descriptive screening estimates at development-level precision; not a validation of the instrument.

Table S3

Period Contrasts on Forum Content Codes: 2023–2026 Versus 2016–2019, With Permutation Tests

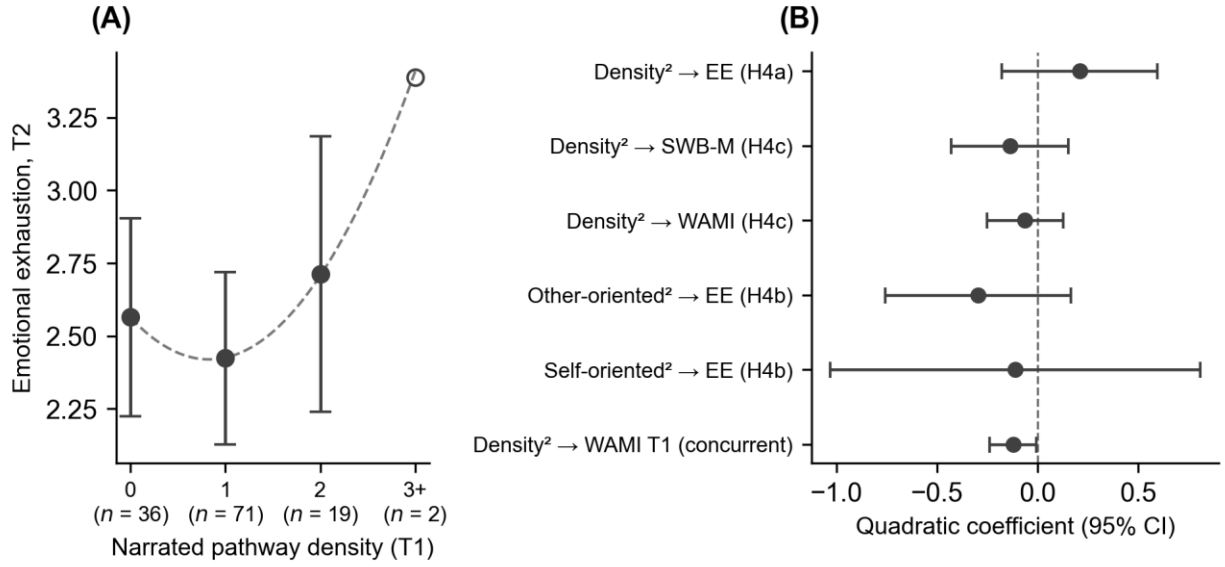
Quantity	2016–2019	2023–2026	Difference [95% CI]	*p*	*q*
Post-level share (% of posts)					
Any optimization, implementation, or AI object	4.8	5.2	0.4 [−1.8, 2.5]	.745	.971
Named AI	0.0	0.0	0.0 [0.0, 0.0]	1.000	1.000
Any frustrated event	16.2	19.8	3.5 [0.2, 6.6]	.034	.138
Frustrated event with such an object	4.0	4.4	0.4 [−1.5, 2.3]	.664	.971
Quit-intent flag	2.2	2.7	0.4 [−0.9, 1.8]	.539	.971
Lack-of-control flag	10.5	14.5	4.0 [1.2, 6.7]	.003	.028
Depressed-affect flag	0.6	0.6	−0.1 [−0.5, 0.5]	.850	.971
Anxious-affect flag	7.6	7.8	0.2 [−1.5, 2.0]	.806	.971
Facet × polarity (events per 100 posts)					
Competence, enacted	11.6	10.7	−0.9 [−3.4, 1.4]	.451	.598
Competence, frustrated	4.8	5.7	0.9 [−0.7, 2.4]	.287	.513
Control/Autonomy, enacted	5.3	6.3	1.1 [−0.6, 2.7]	.213	.513
Control/Autonomy, frustrated	9.6	12.7	3.1 [0.5, 6.0]	.027	.272

Note. Shares are percentages of posts (top block) or events per 100 posts (bottom block); single-model codes. Thread-cluster percentile bootstrap 95% confidence intervals (B = 2,000); p from a two-sided thread-level permutation test of period labels (B = 4,000; threads take their modal period); q = Benjamini–Hochberg within family (eight post-level shares; 20 facet-by-polarity cells with a rate of at least 0.5 per 100 posts, of which the four largest are shown). Against 2020–2022 (not shown), seven enacted facet cells and the two frustration markers rose at $q < .10$, consistent with recovery from the 2019–2021 template-change trough. No category share within object events (19 tests) and no badge stratum (four cells) differed at $q < .10$ in either contrast. The 2020–2022 shares are given in Figures S3–S7. Descriptive screens, not pre-specified tests.

Supplemental Figures

Figure S1

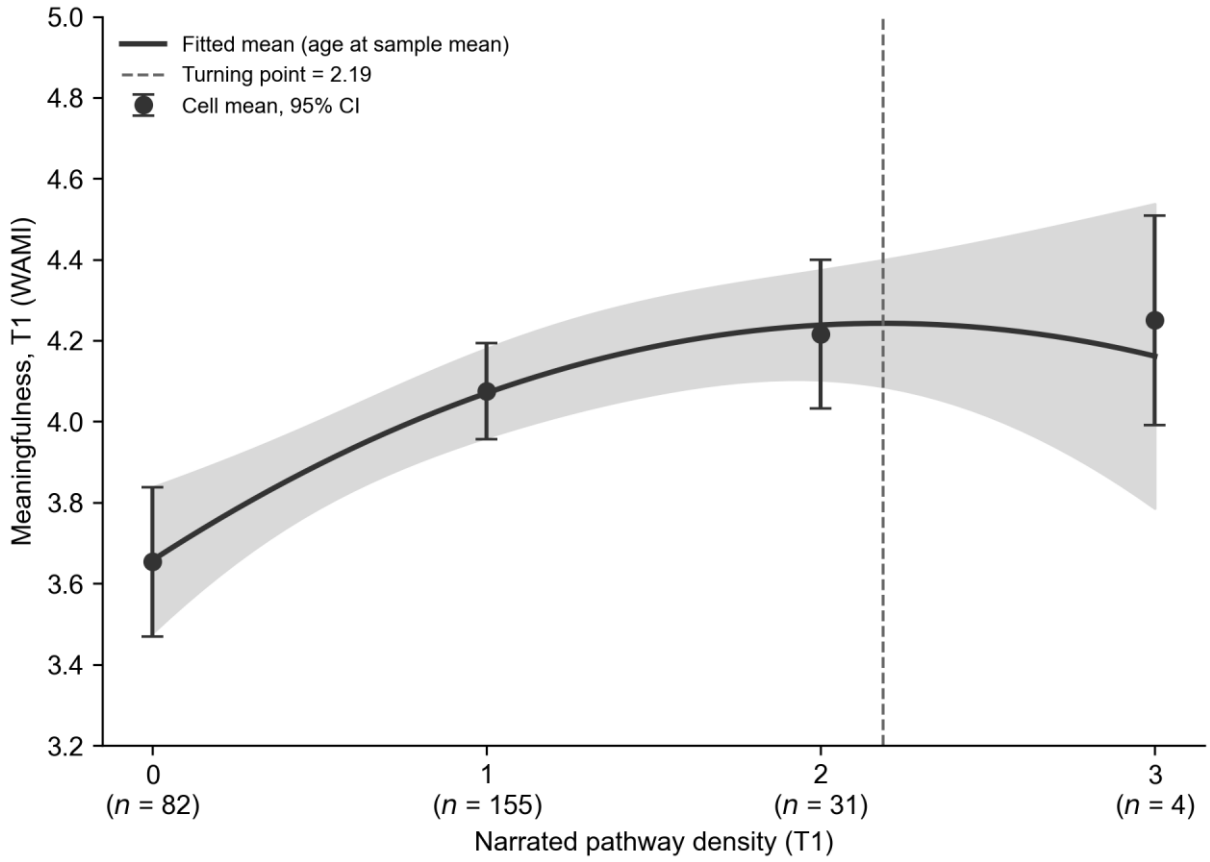
Pathway Density and Time 2 Exhaustion



Note. EE = emotional exhaustion; SWB-M = subjective well-being, meaning of work; WAMI = Work and Meaning Inventory. Panel A: cell means with 95% confidence intervals and the fitted quadratic (age held at the sample mean). The 3+ cell (n = 2) is plotted without an interval. Panel B: quadratic coefficients with 95% confidence intervals. Descriptive only; the sample could not support a formal dark-side test. H4a–H4c label the Block 4 curvilinear hypotheses defined in Figure 1.

Figure S2

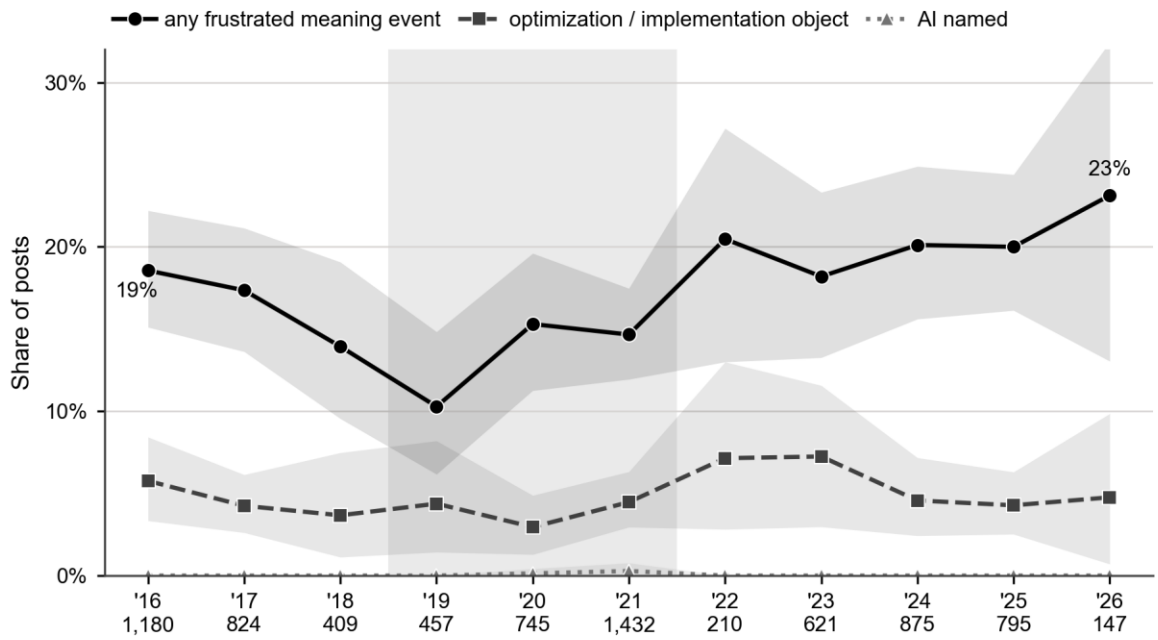
Concurrent Pathway Density and Time 1 Meaningfulness



Note. WAMI = Work and Meaning Inventory; cell sizes sum to $n = 272$. Line is the fitted quadratic with age held at the sample mean; the shaded band is the 95% confidence interval of the fitted mean. Points are observed cell means with 95% confidence intervals. The dashed line marks the turning point (2.19 pathways). Exploratory.

Figure S3

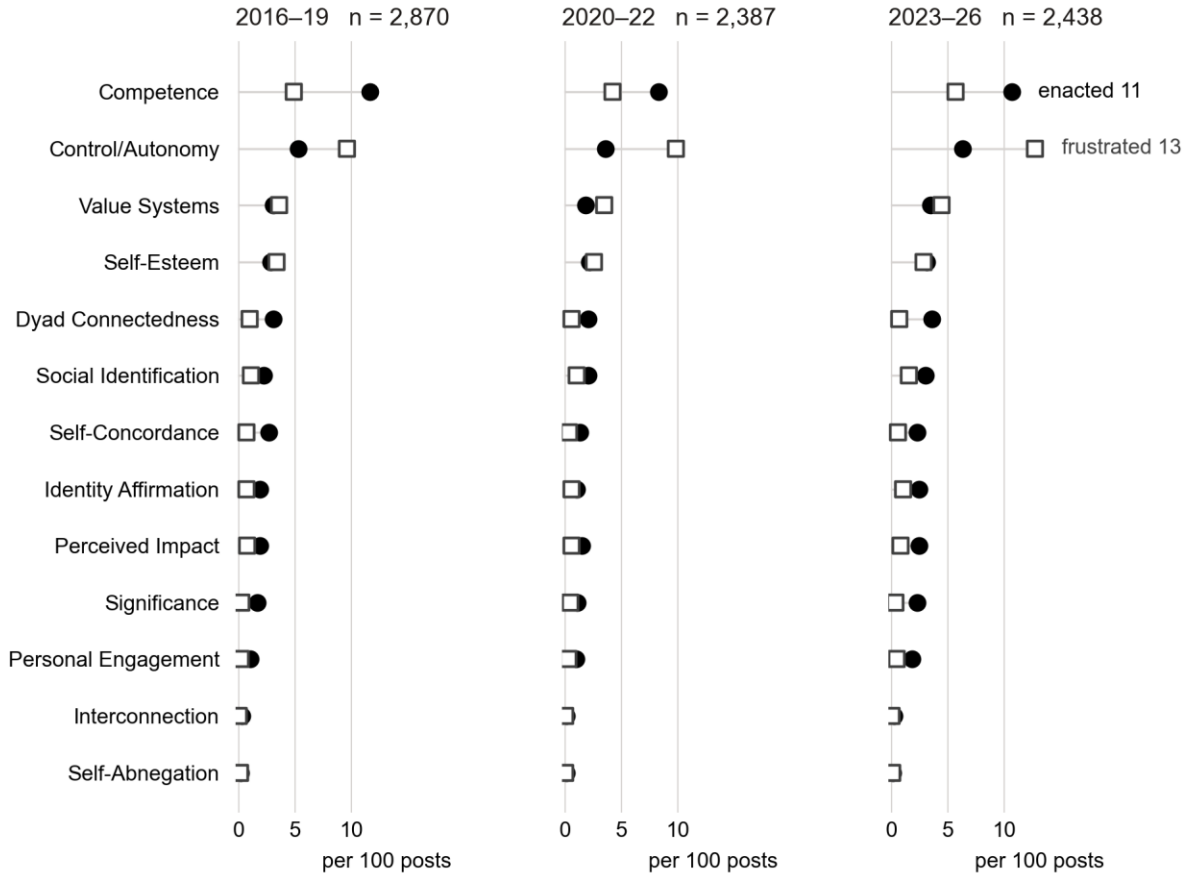
Frustrated Meaning Events, Object Talk, and Named AI by Capture Year



Note. Post-level shares by Wayback capture year with 95% thread-cluster bootstrap intervals (shaded bands); post counts per year appear beneath the axis. The shaded vertical band marks the 2019–2021 page-template change (the site moved from vBulletin to Invision Power Services [IPS] software) that shortened posts and reduced codable events per post. Object = any event whose object is a named artificial intelligence (AI) system, a system implementation, or an optimization practice. Named-AI events: seven in total, none after 2021. Capture year is a proxy for posting year.

Figure S4

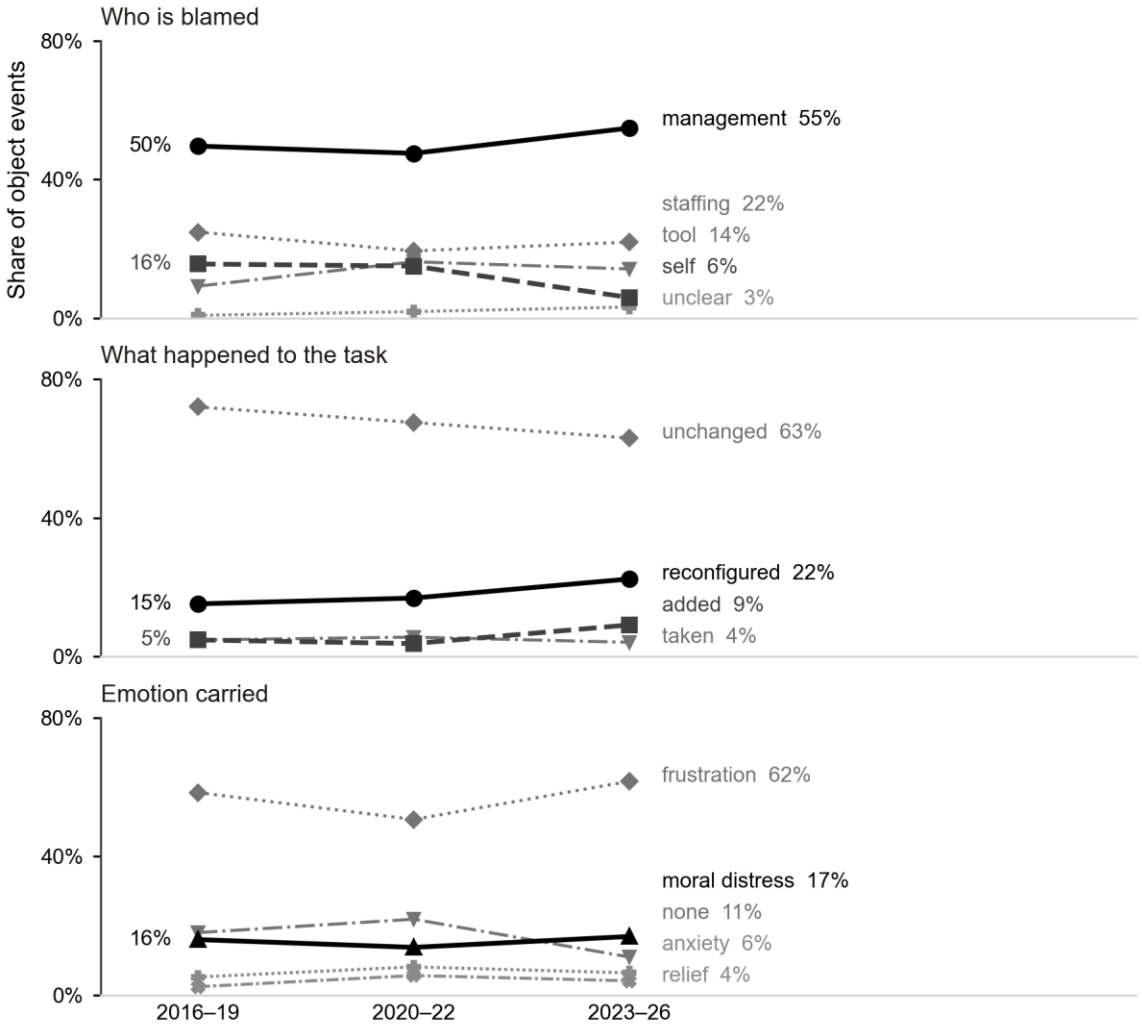
Meaning Events per 100 Posts by Facet, Polarity, and Capture Period



Note. Meaning events per 100 posts by Rosso et al. (2010) facet and capture period (2016–2019, 2020–2022, 2023–2026); single-model codes. Filled circles are enacted events; open squares are frustrated events. Wished-for events (three or fewer per 100 posts) are omitted for legibility. Table S3 reports the tested period contrasts, not every facet. *n* = posts in the period.

Figure S5

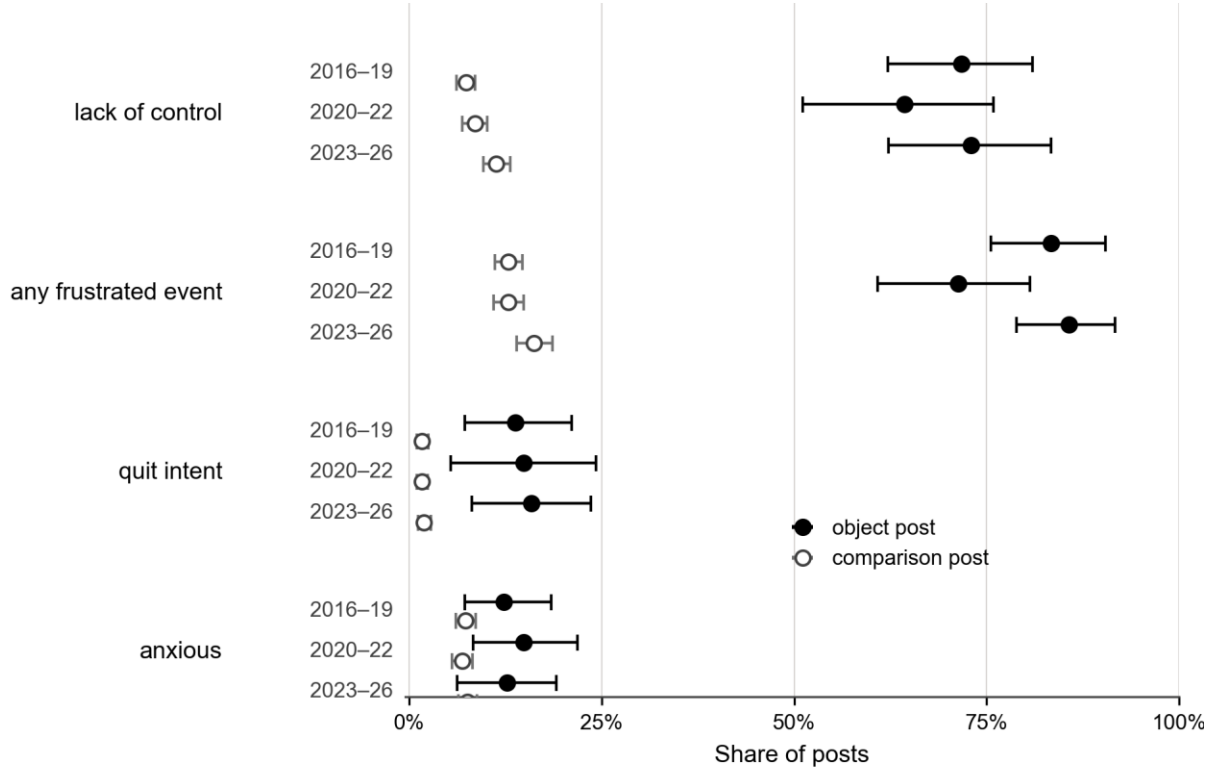
Composition of Object Events: Attribution, Task Outcome, and Emotion by Capture Period



Note. Shares of the 629 object events in each category by period; labels give the 2023–2026 share. Emphasized lines mark the categories that move (management and self attribution; reconfigured and added task outcomes; moral distress). Cells are about 200 events per period, so shifts of five percentage points are within sampling error. Patient attribution (0%) is omitted.

Figure S6

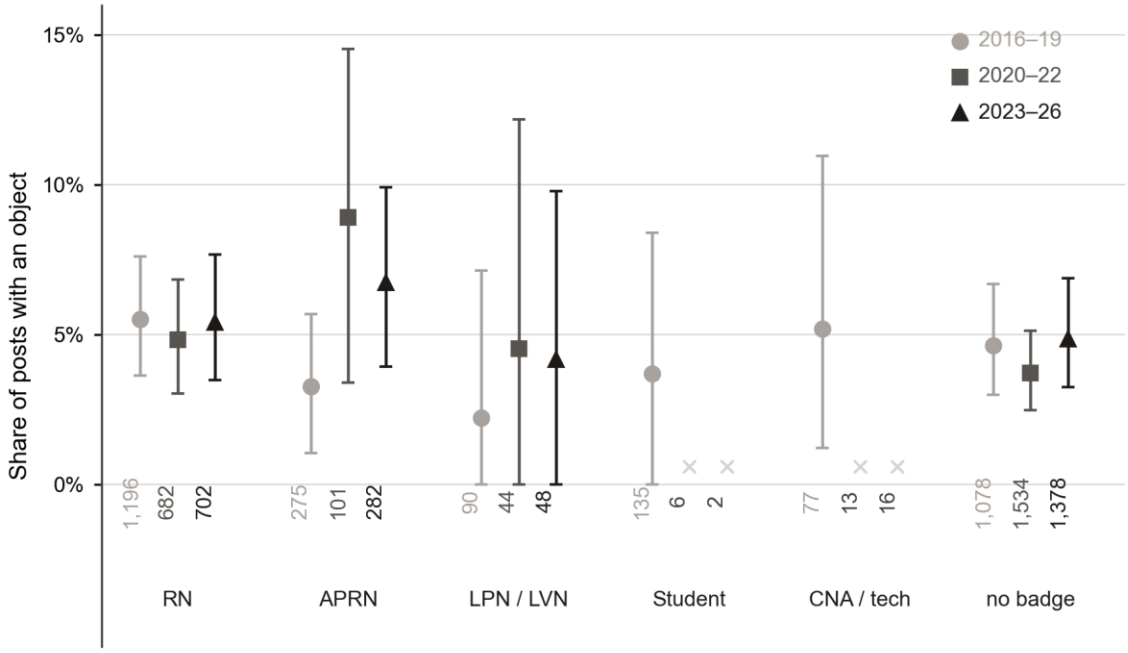
Same-Post Co-Occurrence of Object Talk With Outcome Flags



Note. *P*(flag | object post) versus *P*(flag | other post) within period, 95% thread-cluster bootstrap intervals. One coder, one post; the codebook lets the lack-of-control flag overlap a Control frustration on the same passage, and object posts are complaint posts by construction. Object posts *n* = 138, 101, and 126 per period; comparison posts *n* = 2,732, 2,286, and 2,312. Reverse conditionals: 30% of quit-flagged and 28% of lack-of-control posts named an object. Not a causal or temporal contrast.

Figure S7

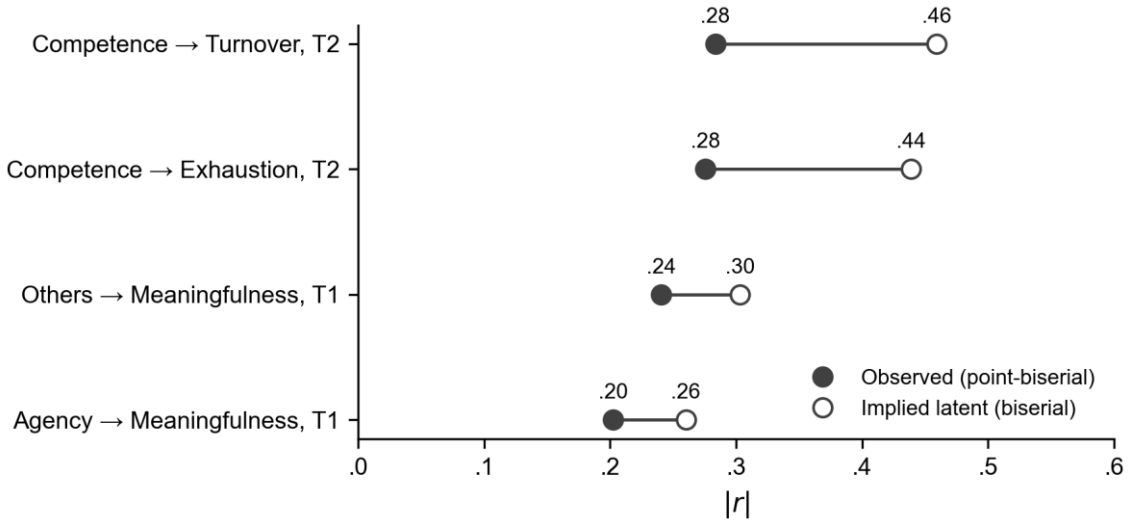
Object Exposure by Career-Point Badge and Capture Period



Note. Share of posts with an optimization or implementation object by author-pane credential badge and capture period (light to dark: 2016–2019, 2020–2022, 2023–2026), with 95% thread-cluster intervals; post counts appear under each marker. Cells with fewer than 40 posts are marked × and not estimated. RN = registered nurse; APRN = advanced practice registered nurse; LPN/LVN = licensed practical or vocational nurse; CNA = certified nursing assistant. Badge is read from the author pane at capture, never from post text; 48% of posts carried a classifiable badge, and coverage depends on page template. The six classes shown cover 7,659 of the 7,695 posts; 36 posts with an educator, management, or other licensed badge are omitted.

Figure S8

Attenuation of Binary-Code Associations



Note. Filled markers are observed point-biserial correlations (all cases with the Time 2 outcome, n = 140 and 132 for turnover intentions and exhaustion; n = 272 for Time 1 meaningfulness); open markers are implied biserial correlations under a normal liability model. Illustrative; latent normality is assumed. $|r|$ = absolute value of the correlation.